UNITARES: Information-Theoretic Governance of Heterogeneous Agent Fleets

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Abstract

UNITARES is a runtime governance layer for heterogeneous AI-agent fleets. It tracks continuous agent state, calibrates by class, detects drift, and issues governance interventions with auditable provenance. Each agent carries a four-dimensional state vector $\mathbf{x} = (E, I, S, V)^\top$ updated on every check-in, intended to support continuous governance rather than discrete after-the-fact filtering. Fleet-wide normalization is the wrong target: the populations we govern—embodied agents with sensor-driven state, persistent autonomous services, session-bounded coding assistants, and ephemeral parser agents—do not share an output modality, a tempo, or a healthy operating point.

UNITARES addresses this with two structural moves. First, it reinterprets the EISV coordinates in information-theoretic terms: S as response-distribution entropy, I as context-response mutual information, E as negative variational free energy, and V as an accumulated free-energy residual. Second, it makes normalization class-conditional: scale constants and healthy operating points are functions of an agent class keyed on existing identity tags, with fleet-wide defaults retained only as fallback behavior.

Because UNITARES was already deployed, regrounding the coordinates raised a live-systems problem in addition to a mathematical one. We introduce a pipeline-ordering migration mechanism that separates decision-producing from presentation-producing pipeline stages so a deployed governance framework can acquire new coordinate semantics without breaking live behavior; we argue this mechanism generalizes to any deployed agent framework facing a semantic-coordinate change. Stability of the dynamics is preserved under the reformulation and established by contraction analysis (Section B).

We illustrate the framework with a production deployment snapshot from February 20, 2026, covering 903 registered agents and 75 active agents. A verdict counterfactual on a 30-day production slice (N = 13,310) shows that 28.9% of basin assignments would flip under class-conditional grounded coherence, with flip rates ranging 15–33% per class and a strong directional bias into the `low` basin, showing that the homogenization-failure-mode argument has first-order consequences at the gating layer rather than only at the reported-value layer. A first per-class calibration of the manifold-coherence path is reported here; the remaining scale constants and higher-tier estimators (logprob and multi-sample) are deferred to subsequent work.

Keywords: autonomous governance, information theory, variational free energy, heterogeneous multi-agent systems, class-conditional calibration, AI safety

Reading aids. We introduce four terms up front: *Variational free energy* ($-F$, used as the E coordinate): the Free-Energy-Principle quantity measuring how well an agent's predictions match outcomes; high E = low surprise. *Cosmological soup*: the failure mode of averaging a heterogeneous agent fleet into a single state distribution, in which no population structure survives normalization. *Manifold coherence* ($C = 1 - \|\Delta\|_2 / \|\Delta\|_{\max}$): distance from the agent class's healthy operating point, used as a class-conditional replacement for the legacy tanh-of-V coherence. *Basin flip*: a change in governance-basin assignment (`high` / `boundary` / `low`) when the same state vector is classified under different coherence formulas.

1 Introduction

Autonomous AI agents operating in production environments face a fundamental tension: they must explore to learn, yet remain safe while doing so. Existing approaches to agent governance fall into two camps. *Hard constraints* (guardrails, output filters) provide safety but impede capability. *Soft monitoring* (logging, dashboards) preserves capability but offers no formal guarantees.

We argue that governance should be *intrinsic*: a continuous dynamical process that shapes agent behavior from within, producing leading indicators of degradation rather than binary verdicts after the fact. This paper presents UNITARES, a framework in which each agent carries a four-dimensional state $\mathbf{x} = (E, I, S, V)^\top$ whose evolution is governed by coupled differential equations. The state provides a continuous, interpretable signal for governance decisions—when to proceed, when to pause, when to escalate.

UNITARES is a deployed framework; the present formulation is organized around two structural commitments: information-theoretic grounding of the EISV coordinates, and class-conditional calibration that treats heterogeneity as a first-class design constraint. The populations the framework governs are heterogeneous—they differ in tempo by four orders of magnitude, in output modality (text, sensor, system action), and in healthy operating point. A governance framework that collapses this population to a single state distribution—what we call *cosmological soup*—loses signal exactly where signal matters. The contributions below are organized around these two commitments.

1.1 Contributions

This paper makes five contributions:

1. **Framework reframing for heterogeneous fleets** (Section 2): We identify heterogeneity, rather than single-population stability, as the governing design constraint for a production agent-governance framework.

2. **Information-theoretic grounding of EISV** (Section 4): We give each coordinate computable information-theoretic semantics, together with tiered estimation recipes and provenance tags.

3. **Class-conditional calibration protocol** (Section 8): We formalize scale constants and healthy operating points as class-indexed quantities derived from identity structure, with explicit fallback behavior for sparse or unclassified classes.

4. **Behavior-preserving migration mechanism for deployed systems** (Section 12): We introduce a pipeline-ordering strategy that separates decision-producing and presentation-producing stages so semantic re-grounding can occur without changing live governance behavior during migration.

5. **Production deployment snapshot and verdict counterfactual** (Section 11): We report a production snapshot of the deployed system to characterize operating regimes, saturation behavior, and observed fleet dynamics. We also report a verdict counterfactual on a 13,310-row production slice showing that 28.9% of basin assignments would flip under class-conditional grounded coherence, with the dominant direction of flips (predominantly into the `low` basin) showing that the homogenization-failure-mode argument of contribution 1 has

first-order consequences at the gating layer, not only at the reported-value layer.

1.2 Limitations and Scope

This paper is a framework-and-migration paper, not a completed empirical validation paper. Its primary claims are architectural and methodological: that heterogeneous fleets require class-conditional calibration, that the EISV coordinates can be reinterpreted information-theoretically without changing the governing dynamics, and that a deployed governance system can be re-grounded by separating decision-producing from presentation-producing stages in the request pipeline.

Accordingly, three boundaries matter. First, the class-conditional calibration protocol is specified here, and a first measurement populating the manifold-coherence path's per-class constants is reported in Section 11.5; the remaining $S_{\text{scale}}, I_{\text{scale}}, E_{\text{scale}}$ constants (used only by the higher-fidelity tier-1 and tier-2 estimators) remain placeholders awaiting those estimators landing. Second, the highest-fidelity estimators for grounded quantities (logprob-based and multi-sample computation) remain future instrumentation work; the current deployment relies on degraded-mode fallbacks where those signals are unavailable. Third, the deployment results in Section 11 are presented as a single descriptive production snapshot rather than a controlled validation study across stable parameter regimes.

The paper should therefore be read as establishing the framework, the migration mechanism, and the shape of the empirical program, while deferring full class-conditional validation to subsequent work.

1.3 Related Work

Information-Theoretic Monitoring. Shannon entropy and mutual information Shannon [1948] are standard tools for characterizing model output distributions; calibration analysis Guo et al. [2017] treats the related question of confidence-outcome alignment. The variational free-energy principle Friston [2010] provides a unifying frame in which agent surprise corresponds to free energy and adaptive agents minimize expected surprise; the active-inference literature Da Costa et al. [2020], Sajid et al. [2021], Smith et al. [2022], Parr et al. [2022] gives a modern treatment of that lineage. We take from this literature the semantic referent for the E coordinate (the variational free-energy concept of Friston [2010]); we do not apply the discrete-state-space machinery of the active-inference treatments, because the agents under governance do not generally expose discrete latent state-spaces to the governance layer. UNITARES adapts the remaining information-theoretic quantities (Shannon entropy, mutual information) to the governance setting and treats the choice of computation tier (logprob, multi-sample, heuristic) as a deployment configuration rather than an assumption.

Contraction Theory. Lohmiller and Slotine Lohmiller and Slotine [1998] established contraction analysis as a tool for nonlinear stability, later extended to Riemannian metrics Slotine [2003]. We adapt this framework to the governance domain, where the "plant" is an agent's state vector rather than a physical system. Contraction is preserved by the grounded reformulation but is no longer the headline contribution; see Section B.

AI Safety and Governance. Constitutional AI Bai et al. [2022] and RLHF Ouyang et al. [2022] govern model outputs through training objectives, while output filters and guardrails Rebedea et al. [2023] enforce constraints at inference time. These approaches treat safety as a discrete property (safe/unsafe) applied at the boundary of agent action. UNITARES instead models safety as a continuous dynamical state that evolves between actions, providing leading indicators of degradation rather than binary verdicts after the fact.

Ravindran Ravindran [2025] proposes a Moral Anchor System (MAS) that detects value drift using a three-dimensional state vector with Bayesian inference and LSTM-based forecasting. The differentiation is twofold. First, *scope*: MAS targets *value*-level drift against a declared value specification, while UNITARES's behavioral–epistemic drift vector (Section 7) explicitly does not measure values—the four components instrument decision consistency, coherence departure, calibration gap, and complexity self-estimation error. The two vectors are complementary channels, not competing formulations. Second, *coupling*: MAS is a statistical detection pipeline that predicts drift from observed behavior; UNITARES couples drift into the state equations themselves via the $\|\Delta\boldsymbol{\eta}\|$ term in Equations (1) and (3), so drift acts as a forcing on the governed dynamics rather than as an external signal.

Concurrent work on runtime governance for agentic systems includes MI9 Wang et al. [2025a], which combines an agency-risk index with finite-state-machine conformance engines and goal-conditioned drift detection. UNITARES differs in representation: where MI9 tracks conformance to an FSM, UNITARES maintains a continuous four-dimensional dynamical state whose evolution itself carries the governance signal. ProbGuard Wang et al. [2025b] takes a probabilistic model-checking route, modeling agent behavior as a discrete-time Markov chain and verifying safety properties against a PRISM specification; UNITARES operates in a continuous state space with no external property specification, and its governance signal is the state itself rather than a conformance verdict. Rath Rath [2026] independently proposes an Agent Stability Index over multi-agent LLM systems as a composite metric; UNITARES's behavioral–epistemic drift vector (Section 7) couples drift back into the governing equations rather than tracking it as an external metric.

Control Barrier Functions (CBFs) Ames et al. [2019] enforce forward invariance of safe sets for multi-agent systems. CBFs guarantee that the system remains within a safe region but do not provide convergence to a desired operating point. Contraction theory, by contrast, guarantees exponential convergence to a unique trajectory, a strictly stronger property that subsumes set invariance when the equilibrium lies within the safe set.

Adaptive Control. PID control is classical Åström and Murray [2008]; our contribution is applying it to governance thresholds with phase-aware reference tracking, connecting it to the EISV state vector through the coherence function.

Calibration. Confidence calibration has a rich literature in forecasting and ML Guo et al. [2017], Niculescu-Mizil and Caruana [2005]. Our behavioral–epistemic drift vector uses calibration deviation as one of four measurable components, connecting calibration to governance dynamics.

2 Heterogeneous Agent Fleets

A central design constraint of UNITARES is that the population it governs is heterogeneous. The deployments that motivated the framework include an embodied creature with I^2C sensors sampled at sub-Hertz rates, a cron-driven janitorial agent that wakes once every 30 minutes, an analytical observer that streams continuously over a websocket, a session-bounded coding assistant generating text at 40–80 tokens per second, and ephemeral parser agents whose entire lifecycle measures in milliseconds. These do not share a tempo. Their outputs are not in the same units. They do not have a common healthy operating point. A governance framework that collapses them to a single state distribution loses signal exactly where signal matters.

2.1 The Homogenization Failure Mode

Most agent-monitoring frameworks assume a single inference layer producing comparable outputs at comparable rates—typically a single LLM behind a single client API. Under that assumption, a fleet-wide normalization is harmless: every agent's response distribution lives on the same support; every agent's expected throughput sits in the same envelope; a single percentile of a single quantity describes the whole population.

UNITARES violates each of these assumptions by construction. The framework runs governance check-ins from agents whose outputs are not text (sensor readings; system actions; drawing strokes), whose tempos differ by four orders of magnitude (continuous streams to days-between-wakes), and whose "healthy operating point" is genuinely class-dependent. A divergent/exploratory agent operating at high entropy is doing exactly what we want; flagging it as incoherent against an analytical-class baseline is not a governance signal but a category error.

We refer to the failure mode of fleet-wide normalization as *cosmological soup*: a state distribution that is maximally entropic, with no class structure and no distinguishability between populations whose dynamics are genuinely different. The information-theoretic grounding presented in Section 4 resolves the metaphor problem in the coordinates themselves; the class-conditional calibration presented in Section 8 resolves the homogenization problem at the normalization layer. These are distinct contributions and either can fail independently.

2.2 Class Structure in the Identity Layer

UNITARES already carries the keying material for class assignment. Every agent registered in the system has an identity record with two relevant fields: a set of *tags* drawn from a controlled vocabulary, and an optional *label* (a human-readable name). The tags {`embodied`, `autonomous`, `persistent`, `ephemeral`, `pioneer`} partition the population along axes that are governance-relevant: embodiment determines whether the output channel is text or sensor-data; autonomy determines whether the agent self-directs or responds to prompts; persistence determines whether the trajectory accumulates over days or resets every session.

Table 1 presents the live deployment classes that motivate the v6 framework, drawn from production identity records.

The natural calibration partition is along these tags and labels: distinct scale constants and healthy baseline points per class, with the fleet-wide constant retained as a fallback for unclassified agents. Phase 2 calibration formalizes this partition (see Section 8); the present section establishes the requirement.

2.3 Why Heterogeneity Is the Differentiator

The contraction-theoretic stability result (Section B) demonstrates that the EISV dynamics are well-behaved under standard control-theoretic conditions. That result is not the headline contribution. Stability of the dynamics is a property of any reasonable governance equation of this form; what distinguishes UNITARES is that the same dynamics describe a heterogeneous population without collapsing the classes that define them. The present framing makes that distinction explicit: the contribution is the class-conditional grounding, not the dynamics-on-a-single-class.

3 EISV Dynamics

3.1 State Space

The UNITARES state is a four-dimensional vector $\mathbf{x} = (E, I, S, V)^\top$ evolving on the compact set

$$\mathcal{X} = [0,1]^3 \times [-1,1]$$

Each coordinate is a normalized information-theoretic quantity; the formal definitions appear in Section 4, with class-conditional normalization in Section 8. This section presents the dynamics of the coordinates as a self-contained dynamical system, deferring the grounding of each coordinate to Section 4.

Definition 3.1 (EISV State Variables)**.**

- $E \in [0,1]$*: **Energy**—negative variational free energy of the agent, normalized to* $[0,1]$*. High E corresponds to low surprise on outcomes (the agent's predictions match what happens).* Grounded definition: *Equation* (10).
- $I \in [0,1]$*: **Information integrity**—mutual information between context and response,*

Table 1: Live agent classes in the production deployment.

Agent	Tags	Output	Tempo	Persistence
Lumen	embodied, persistent	sensor + display	0.5 Hz	continuous
Vigil	autonomous, persistent	shell + audit	1/30 min	cron
Sentinel	autonomous, persistent	websocket	~ 1 Hz	continuous
Watcher	autonomous, ephemeral	finding records	event-driven	per-edit
Steward	autonomous, persistent	sensor sync	1/5 min	continuous
Claude / Codex sessions	—	text response	40–80 tok/s	session-bounded
Parser agents	ephemeral	structured text	burst	ms-lifetime

normalized. Grounded definition: *Equation* (9)*.*

- $S \in [0,1]$*: **Entropy**—Shannon entropy of the agent's response distribution, normalized via a class-conditional scale constant. Bounded below by an epistemic humility floor* $S_{\min} = 0.001$*.* Grounded definition: *Equation* (8)*.*

- $V \in [-1,1]$*: **Void / Free-Energy Debt**—the exponentially-weighted accumulated free-energy residual. Structurally identical to a damped accumulator of the E−I gap; under the grounded interpretation it tracks departure from the agent's rolling free-energy baseline.* $V > 0$ *indicates running "hot" (higher surprise than baseline);* $V < 0$ *indicates running "careful" (integrity exceeding energy).* Grounded definition: *Equation* (11)*.*

Remark 3.2 (State-space bounds)**.** *The grounded definitions of Section 4 normalize each coordinate to* $[0,1]$ *(or* $[-1,1]$ *for* V*) by construction via the scale constants of Section 8; production data confirms the normalization acts as intended, with* V *and* S *observed two orders of magnitude inside the nominal bounds (Section 11).*

3.2 Governing Equations

The dynamics are:

$$\frac{dE}{dt} = \alpha(I - E) - \beta_E ES + \gamma_E \|\Delta\boldsymbol{\eta}\|^2 \tag{1}$$

$$\frac{dI}{dt} = -kS + \beta_I C(V,\Theta) - g_I(I) \tag{2}$$

$$\frac{dS}{dt} = -\mu S + \lambda_1(\Theta)\|\Delta\boldsymbol{\eta}\|^2 - \lambda_2(\Theta)C(V,\Theta) + \beta_c\mathcal{C} + \sigma\xi(t) \tag{3}$$

$$\frac{dV}{dt} = \kappa(E - I) - \delta V \tag{4}$$

where $g_I(I) = \gamma_I I$ is the information integrity self-regulation term, $C(V,\Theta)$ is the coherence function (Section 3.3), $\|\Delta\boldsymbol{\eta}\|$ is the behavioral–epistemic drift norm (Section 7), $\mathcal{C} \in [0,1]$ is task complexity, and $\xi(t)$ is a noise process with intensity σ. A logistic alternative $g_I^{\log}(I) = \gamma_I I(1 - I)$ has a boundary-saturation pathology that the linear form avoids; the analysis is collected in Section C.

Remark 3.3 (Coupling structure under the grounded interpretation)**.** *The coupling structure carries direct information-theoretic meaning:* E *(precision / negative free energy) tracks* I *(mutual information with context)—the agent becomes more confident when its context is more informative;* I *is degraded by entropy* S *(response-distribution uncertainty) but restored by coherence (proximity to the calibrated healthy operating point);* S *relaxes toward its prior under stationary conditions* ($-\mu S$) *and is driven up by behavioral–epistemic drift and task complexity;* V *accumulates the* $E-I$ *residual with damping* δ*, which under the grounded interpretation is the* *exponentially-weighted free-energy debt of Equation* (11).

3.3 Coherence Function (Operational Form)

UNITARES maintains two coherence presentations: a *grounded form* (manifold distance from the healthy operating point, Equation (13); or KL divergence from a reference distribution, Equation (12)) used in the response payload, and a legacy *operational form* that feeds back into the dynamics through (2) and (3). The operational form is

$$C(V, \Theta) = C_{\max} \cdot \tfrac{1}{2}\left(1 + \tanh(C_1 \cdot V)\right) \tag{5}$$

with $C_{\max} = 1$ and $C_1 = 1.0$ (production values). This maps $V \in \mathbb{R} \to [0, C_{\max}]$ with the steepest sensitivity at $V = 0$. The dynamics consume the operational form; the grounded form is reported in canonical payload slots. Section 12 documents the retirement sequence for the operational form.

Remark 3.4 (Operating Range)**.** *In production, the damping term* δV *in* (4) *constrains* V *to approximately* $[-0.1, 0.1]$*, yielding* $C(V) \approx 0.49$*; the empirical distribution is reported in Section 11. The narrow range near the coherence-function midpoint is the operational consequence of the damping analysis rather than a deficiency of the signal.*

3.4 Adaptive Entropy Coupling

The entropy source term $\lambda_1(\Theta)$ is adaptive, coupling the governance parameter vector Θ to the dynamics:

$$\lambda_1(\Theta) = \lambda_{1,\min} + \frac{\eta_1 - \eta_{1,\min}}{\eta_{1,\max} - \eta_{1,\min}} \cdot (\lambda_{1,\max} - \lambda_{1,\min}) \tag{6}$$

where $\eta_1 \in [0.1, 0.5]$ maps to $\lambda_1 \in [0.05, 0.20]$. Higher drift sensitivity increases entropy injection, creating a natural feedback loop: drifting agents become more uncertain, which in turn triggers conservative governance decisions.

3.5 Objective Function

The scalar governance signal is:

$$\Phi = w_E E - w_I(1-I) - w_S S - w_V\,|V| - w_\eta\,\|\Delta\boldsymbol{\eta}\|^2 \tag{7}$$

with default weights $w_E = w_I = w_S = w_V = w_\eta = 0.5$. Governance verdicts follow thresholds: $\Phi \geq 0.15$ (proceed), $0 \leq \Phi < 0.15$ (caution), $\Phi < 0$ (high-risk/pause).

Remark 3.5. *The threshold* $\Phi = 0.15$ *is calibrated empirically: a healthy agent with* $E = 0.7$*,* $I = 0.8$*,* $S = 0.2$ *yields* $\Phi \approx 0.15$*. These values are illustrative; per-class healthy operating points (Section 11.5) refine this calibration in deployment.*

4 Information-Theoretic Grounding

The $\mathbf{x}$ system presented in Section 3 specifies a coupled dynamical signal for governance decisions but does not specify what each coordinate measures. This section supplies the information-theoretic semantics: each coordinate becomes a computable quantity with a tiered estimation recipe and a provenance tag, without changing the dynamics of Section 3.2.

Remark 4.1 (What stays, what changes)**.** *The ODE structure in Equations* (1) *to* (4)*, the contraction analysis in Section 5, the governor in Section 6, and the verdict thresholds in Equation* (7) *are all preserved. Grounding changes what* E*,* I*,* S*, and* $C(V, \Theta)$ are*; it does not change what the governance surface* does.

4.1 Choice of Frame

We adopt a Shannon plus variational (free-energy) hybrid. Shannon information theory gives genuinely computable entropy and mutual information from agent outputs—data already captured by the tool usage recorder and the governance audit log. The Free Energy Principle Friston [2010] supplies the *target* formalization: under FEP, agent drift corresponds to rising variational free energy, and free-energy monitoring is the asymptote toward which governance aims. Tier-1 instrumentation of the agent's generative model is not currently available for closed-weights frontier agents; until it is, the deployed system uses provenance-tagged proxies for each coordinate (Section 4.2) and flags each channel's tier on a per-agent-turn basis, so downstream consumers can read the coordinate alongside its estimator commitment. The FEP inherits a derivation from statistical mechanics through variational Bayes, giving a real ancestry for the free-energy quantity interpreted in V, even when V is currently driven by the proxy form in Equation (11) rather than a Tier-1 residual.

Alternatives considered and rejected: pure control-theoretic Lyapunov framing (abandons the FEP/free-energy lineage entirely); information geometry with Fisher metric (more machinery than the framework needs at this stage); pure statistical mechanics with a partition function (requires defining a Hamiltonian analog for agent behavior, which is itself arbitrary at the choice of Hamiltonian).

The resulting correspondence between each coordinate and its information-theoretic referent is:

Information / FEP quantity	**Coord.**
Shannon entropy of $q(y \mid x)$	S
Mutual information $\mathrm{MI}(x; y)$	I
Negative variational free energy $-F$	E
Accumulated free-energy residual	V
KL / manifold distance from healthy	C

The symbols $\{E, I, S, V, C\}$ are retained for continuity with deployed systems; their semantic load is now the right column.

4.2 Quantity Redefinitions

Each coordinate of $\mathbf{x}$ receives (a) a formal information-theoretic definition, (b) a tiered computation recipe in order of preference, and (c) an explicit data dependency.

S—**Entropy.** Let $q(y \mid x)$ denote the agent's distribution over responses y given context x at a given turn. Define

$$S_{\mathrm{raw}} = H(q) = -\sum_y q(y \mid x) \log q(y \mid x),$$
$$S = 1 - \exp(-S_{\mathrm{raw}}/S_{\mathrm{scale}}) \tag{8}$$

in nats, normalized to $[0, 1]$ by a scale constant S_{scale}. Tier 1: compute $H(q)$ from per-token logprobs when the inference layer exposes them. Tier 2: semantic entropy Kuhn et al. [2023]—draw k responses at temperature $T > 0$, group by semantic equivalence, and compute the entropy of the equivalence-class distribution as an estimator of $H(q)$. Tier 3: retain the legacy heuristic (complexity, behavioral–epistemic drift) as a degraded-mode fallback tagged `s_source = "heuristic"` so downstream consumers know the channel is uncalibrated.

I—**Information Integrity.** Under Shannon, I is the mutual information between context and response,

$$I_{\mathrm{raw}} = \mathrm{MI}(x; y) = \mathbb{E}_{q(x,y)}\left[\log \frac{q(y|x)}{q(y)}\right],$$
$$I = \mathrm{clip}(I_{\mathrm{raw}}/I_{\mathrm{scale}},\ 0,\ 1) \tag{9}$$

Tier 1 uses paired logprobs (context-only vs. context+response). When the marginal $q(y)$ is unavailable, a fallback is the KL divergence from a context-free reference distribution $q_{\mathrm{ref}}(y)$: $I = 1 - \min(1, D_{\mathrm{KL}}(q(y \mid x) \,\|\, q_{\mathrm{ref}}(y))/I_{\mathrm{scale}})$. This KL fallback is a divergence from a reference distribution, not $\mathrm{MI}(x; y)$, and is tagged `i_source = "kl_fallback"` so downstream consumers know the channel is a reference-distribution proxy rather than a direct MI estimate. Tier 3 retains the legacy logistic heuristic.

E—**Negative Free Energy.** Under the variational frame,

$$F = \mathbb{E}_q[\log q(z) - \log p(z, o)],$$
$$E_{\mathrm{raw}} = -F, \qquad E = \sigma(E_{\mathrm{raw}}/E_{\mathrm{scale}}) \tag{10}$$

where $q(z)$ is the agent's approximate posterior, $p(z, o)$ its generative model over latents and outcomes, and σ the logistic function. Intuitively,

E is high when the agent's predictions match outcomes (low surprise) and low otherwise. Full variational computation of $-F$ requires instrumentation that exposes the agent's generative model and is deferred to subsequent work. In the current deployment, a *resource-rate* heuristic proxy $E_{\text{resource}} = (\text{tokens}/s)/(\text{tokens}/s)_{\max}$ is used where available, directly measurable from inference metadata. The resource-rate form is *not* equivalent to $-F$ and does not approximate it under stationarity in any formal sense; it is a throughput heuristic tagged `e_source = "resource"` so downstream consumers know the channel is an uncalibrated proxy. The `e_source` annotation distinguishes genuine variational $-F$ from the resource-rate proxy on a per-agent-turn basis.

V**—Free-Energy Debt.** The existing V accumulator in Equation (4) is structurally an exponentially-weighted residual; under the FEP reinterpretation it is the accumulated free-energy residual

$$V_t = \lambda V_{t-1} + (1-\lambda)(F_t - F_{\text{ref}}) \qquad (11)$$

with F_{ref} the agent's rolling baseline free energy. The interpretation shifts—V is now explicitly free-energy debt rather than a phenomenological "void"—while the arithmetic is unchanged. No payload or threshold change is required.

C**—Coherence.** We replace the hand-shaped tanh in Equation (5) with one of two equivalent grounded forms. The information-theoretic form is

$$C = \exp(-D_{\text{KL}}(q_{\text{now}} \,\|\, q_{\text{ref}})) \qquad (12)$$

where q_{now} is a sliding-window response distribution and q_{ref} is a calibrated healthy baseline. The equivalent manifold (Lyapunov) form is

$$\begin{aligned} C &= 1 - \|\Delta\|_2 \,/\, \|\Delta\|_{\max}\,, \\ \Delta &= (E, I, S) - (E, I, S)_{\text{healthy}} \end{aligned} \qquad (13)$$

The two are equivalent under a Gaussian approximation of the state distribution around the healthy operating point. The manifold form ships first because it requires only the existing $\mathbf{x}$ coordinates; the KL form requires an empirically calibrated reference.

4.3 Scale Constants and Provenance

Every scale constant introduced in Equations (8) to (10) and (13) carries an explicit provenance tag recording whether its value is a *placeholder* (shipped as scaffolding, pending a higher-fidelity estimator), *measured* (fit against a reference corpus of healthy production agent-turns), or *derived* (computed from other measured constants). For each constant $\kappa \in \{S_{\text{scale}}, I_{\text{scale}}, E_{\text{scale}}, \|\Delta\|_{\max}\}$, the implementation records the numerical value, the ISO date on which it was set, the corpus size, and the percentile basis (typically 90th or 95th). As of v6, $\|\Delta\|_{\max}$ has been measured per-class (Section 11.5); the remaining constants, which are consumed only by the tier-1 logprob and tier-2 multi-sample estimators, remain placeholders until those estimators land.

4.4 Dual-Compute Migration

The grounded coordinates ship alongside the legacy quantities in a three-phase migration so the deployed governance surface is never broken mid-flight. Section 12 documents the mechanism; for the remainder of this section we treat the grounded definitions as the semantic commitment of the framework.

Remark 4.2 (What the paper's §2 equations mean after grounding)**.** *The equations in Section 3.2 keep their shape. A dE/dt term that read "energy couples toward information integrity" is now "the agent's precision couples toward the informativeness of its context." A dS/dt term that read "entropy decays" is now "the response-distribution entropy relaxes toward its prior under stationary conditions." Cross-couplings become higher-order correlation terms in the variational dynamics. Each ODE coefficient must be labeled with its provenance class (measured, derived, or calibration-required) as of Phase 2; this paper's parameter tables (Section A) will be updated accordingly.*

5 Stability

Contraction Lohmiller and Slotine [1998] means that any two trajectories of the system converge exponentially toward each other under a Riemannian metric M at rate $\alpha_c > 0$; for UNITARES this implies governance trajectories forget their initial conditions and settle onto a unique equilibrium regardless of start state. The EISV dynamics of Section 3 are globally exponentially stable in this sense. Specifically, the system is contracting with rate $\alpha_c > 0$ under the metric $M = \mathrm{diag}(0.1, 0.2, 1.0, 0.08)$, whose diagonal weights reflect the relative scales of the four channels: S fluctuates fastest and is given unit weight, while V is damped most and given the smallest weight (derivation in Section B). The spectral abscissa of the linearized system at the equilibrium gives an actual rate $\alpha_c \approx 0.13$ at production parameters; a conservative Gershgorin argument establishes a strictly-less lower bound $\alpha_c \geq 0.019$ used only to make the stochastic and network bounds in Sections 9 and 10 worst-case conservative. Both certificates appear in Section B.

Formally: under linear information dynamics ($g_I(I) = \gamma_I I$) and production parameters, the EISV system admits a unique equilibrium $\mathbf{x}^*$ to which all trajectories converge exponentially at rate $\alpha_c \geq 0.019$ in the metric M above (Theorem B.2, with full proof in Section B). We deliberately keep the proof in the appendix: stability is a property the dynamics should have, not a contribution that distinguishes UNITARES from related frameworks. Class-conditional grounding (Sections 4 and 8) is the contribution that does.

The grounded reformulation of Section 4 preserves stability *within each agent class*. The dynamics $\dot{\mathbf{x}} = f(\mathbf{x}, t)$ are unchanged; only the meaning of the coordinates and the values of the scale constants change. The contraction certificate of Theorem B.2 relies on fleet-wide bounds (notably $\beta_I C'_{\max}$); under class-conditional normalization, the relevant bounds become class-indexed, and the contraction *structure* carries through class-by-class, with each class acquiring its own $\alpha_c(c)$ upon re-certification at its class-conditional healthy operating point. The equilibrium $\mathbf{x}^*$ becomes the class-conditional healthy operating point of Theorem 8.2 in deployments where calibration has been performed. A per-class numerical rate is not computed in this paper: the deployed certificate of Section B remains valid for the legacy tanh-of-V coherence, and re-certification under a grounded coherence is treated in Theorem 5.1. Cross-class network synchronization (Theorem 10.1) is therefore a within-class property, as already qualified in Section 10; fleet-wide synchronization across heterogeneous classes is not claimed and would not be expected.

Remark 5.1 (Contraction under a grounded coherence)**.** *The contraction proof of Section B consumes the operational* $C(V, \Theta) = \tanh$*-based form through the off-diagonal bound* $\beta_I C'_{\max} = 0.15$. *A deployment that rewires the dynamics to consume a grounded coherence (Equation* (13) *or Equation* (12)*) directly replaces* $\partial C(V, \Theta)/\partial V$ *by derivatives with respect to* (E, I, S)*, altering the off-diagonal bound on which the Gershgorin argument depends. Re-certification at the new operating point with the new coherence derivative is required; the proof structure adapts but the numerical rate* α_c *will change. The migration sequence of Section 12 is designed so this rewiring is the final step, preserving the deployed contraction certificate until it is replaced.*

6 Adaptive Governor (CIRS v2)

Static governance thresholds are fundamentally mismatched to agents that transition between exploration and integration phases. We introduce the Contextually-Informed Regulation System version 2 (CIRS v2), a PID controller that manages governance thresholds as per-agent state variables.

6.1 Architecture

For each agent, CIRS v2 maintains two adaptive thresholds:

- τ: coherence threshold (minimum acceptable coherence)
- β: risk threshold (maximum acceptable risk)

with hard safety bounds $\tau \in [\tau_{\min}, \tau_{\max}]$, $\beta \in [\beta_{\min}, \beta_{\max}]$ that cannot be overridden by adaptation.

6.2 Phase Detection

The agent's operating phase is detected from EISV trajectory trends over a sliding window of w steps:

Definition 6.1 (Phase Detection). *An agent is in* exploration *phase when at least 2 of 3 signals are active:*

1. *Information growth:* $\Delta I/w > \theta_I$ *(*$\theta_I = 0.008$*)*
2. *Entropy decline:* $-\Delta S/w > \theta_S$ *(*$\theta_S = 0.008$*)*
3. *High complexity:* $\bar{C} > \theta_c$ *(*$\theta_c = 0.5$*)*

Otherwise the agent is in integration *phase.*

6.3 PID Control Law

Let $r_\tau^{(p)}$ and $r_\beta^{(p)}$ be the phase-dependent reference points:

$$\text{Exploration:} \quad r_\tau = 0.35,\ r_\beta = 0.55 \tag{14}$$
$$\text{Integration:} \quad r_\tau = 0.40,\ r_\beta = 0.60 \tag{15}$$

The error signals are $e_\tau = r_\tau - \tau$ and $e_\beta = r_\beta - \beta$. The PID update for threshold $\theta \in \{\tau, \beta\}$ is:

$$u_\theta = K_p e_\theta + K_i \int_0^t e_\theta \, dt' + K_d \cdot d_p \cdot \frac{de_\theta}{dt} \tag{16}$$

with gains $K_p = 0.05$, $K_i = 0.005$, $K_d = 0.10$ and phase-dependent derivative factor:

$$d_p = \begin{cases} 0.5 & \text{exploration (tolerate oscillation)} \\ 1.0 & \text{integration (full damping)} \end{cases} \tag{17}$$

Remark 6.2 (The D-term IS the damping). *The derivative gain* $K_d = 0.10$ *is intentionally the strongest term. Oscillation in governance decisions produces large* de/dt*, which produces large corrective updates. The system self-stabilizes without requiring a separate oscillation detector.*

6.4 Integral Wind-Up Protection

The integral term is clamped:

$$\left| \int_0^t e_\theta \, dt' \right| \leq I_{\max} = 0.10 \tag{18}$$

with zero-crossing reset: when $\text{sign}(e_\theta)$ changes, the integral accumulator is zeroed.

6.5 Threshold Decay

When the oscillation index OI < 0.5 (stable), thresholds decay toward defaults at rate $\lambda_d = 0.01$ per update:

$$\theta \leftarrow \theta + \lambda_d(\theta_{\text{default}} - \theta) \tag{19}$$

6.6 Governance Decision

The adaptive governor produces a binary verdict:

$$\text{verdict} = \begin{cases} \texttt{proceed} & C(V) \geq \tau \ \wedge \ R < \beta \\ \texttt{pause} & \text{otherwise} \end{cases} \tag{20}$$

where R is the risk score derived from Φ (Equation (7)).

Remark 6.3 (Why two tiers). *The binary contract is paired with the PID governor's continuous threshold adaptation: the discrete verdict fires or it does not, while the governor carries the nuance through a derivative signal on* τ *and* β*. A middle* `revise` *tier would be operationally indistinguishable from* `proceed` *for agents whose next action is unchanged by the verdict's content, so the nuance is better absorbed by the governor than by an additional verdict level.*

Remark 6.4 (Runtime labels and audit vocabulary). *The binary verdict of Equation* (20) *is the formal contract; deployed layers present refined views that map onto it. Runtime consumers (the agent-facing MCP surface, documentation, and agent-facing skills) expand the verdict into four labels* `proceed` / `guide` / `pause` / `reject`*, where* `guide` *is a* `proceed` *carrying soft advisory text and* `reject` *is a* `pause` *carrying a recovery requirement. The audit schema records a distinct view:* `governance.audit.outcome_events.eisv_verdict` *stores risk bands* **`safe`** */* **`caution`** */* **`high-risk`** *derived from the risk score R of Equation* (7)*, while hard pauses are emitted separately as* **`lifecycle_paused`** *and* **`lifecycle_resumed`** *on* **`audit.events`***. Readers reproducing pause-stream metrics against production data should also filter system tasks such as* **`eisv-sync-task`***, which appear in the stream but are scheduled background processes, not governed agents.*

6.7 Ephemeral Agents and the Sliding Window

CIRS v2 as specified above assumes an agent lives long enough for a sliding window of w steps to populate. Phase detection, integral accumulation, and threshold decay are all well-defined only once w samples exist. Ephemeral classes (Table 1: parser agents with millisecond lifetimes, one-shot tool invocations) cannot satisfy this precondition by construction—the entire agent trajectory is a single step. For these classes the per-agent PID controller degenerates.

The deployed behavior is a class-level rather than per-agent policy: an ephemeral agent inherits (τ, β) directly from its class's stationary reference points $(r_\tau^{(p)}, r_\beta^{(p)})$ with $p =$ integration (the stricter of the two phases, chosen conservatively because no phase signal is available), bypassing the PID loop entirely. The cost is that ephemeral classes do not benefit from per-agent threshold learning; the benefit is that a 5 ms parser is not blocked waiting for a 20-step window to populate. A principled treatment of sub-window-lifetime agents—possibly by pooling the PID state across a class cohort rather than per-agent—is future work.

7 Behavioral–Epistemic Drift Vector

7.1 Motivation and Scope

The behavioral–epistemic drift vector instruments four signals: decision consistency and coherence departure (behavioral), calibration gap and complexity self-estimation error (epistemic). None of them measure the agent's values in a normative sense; the vector is a deliberately behavioral–epistemic quantity. Affective and value-level drift are distinct channels that a complete treatment would add; we discuss those as future work at the end of this section. The remainder of this section grounds the behavioral–epistemic vector in four measurable signals, each bounded in $[0, 1]$.

Remark 7.1 (Information-theoretic re-derivations)**.** *Under the information-theoretic frame of Section 4, two of the four signals have natural re-derivations: η_{cal} (calibration deviation) is a mis-calibration KL divergence between predicted and empirical outcome distributions, and η_{coh} (coherence deviation) is a KL divergence from the class-conditional baseline distribution of Section 4. We use the distance-based forms here; the KL re-derivations are deferred to subsequent work.*

Definition 7.2 (Behavioral–Epistemic Drift Vector)**.**

$$\Delta\boldsymbol{\eta} = \begin{pmatrix} \eta_{cal} \\ \eta_{cpx} \\ \eta_{coh} \\ \eta_{stab} \end{pmatrix} \in [0,1]^4 \tag{21}$$

where:

- η_{cal}*:* ***Calibration****—gap between predicted and actual correctness,* $|\hat{p} - p_{act}|$*.*
- η_{cpx}*:* ***Complexity****—discrepancy between derived and self-reported task complexity,* $|C_{der} - C_{self}|$*.*
- η_{coh}*:* ***Coherence****—departure from EMA baseline,* $|C(V)_t - \bar{C}(V)|$*.*
- η_{stab}*:* ***Stability****—*$1 -$ *consistency over a 20-step decision window.*

7.2 Baseline Tracking

Each component is tracked against an exponential moving average (EMA) baseline with smoothing parameter $\alpha_{\text{ema}} = 0.1$:

$$\bar{x}_{t+1} = (1 - \alpha_{\text{ema}})\bar{x}_t + \alpha_{\text{ema}} x_t \tag{22}$$

A warmup dampening factor prevents spurious drift detection from uninitialized baselines:

$$w(n) = \min\left(\frac{n}{n_{\text{warmup}}}, 1\right), \quad n_{\text{warmup}} = 5 \quad (23)$$

applied as $\Delta\boldsymbol{\eta}_{\text{effective}} = w(n) \cdot \Delta\boldsymbol{\eta}$.

7.3 Coupling to Dynamics

The drift norm $\|\Delta\boldsymbol{\eta}\|$ enters the dynamics at two points:

1. **Energy** (1): $+\gamma_E \|\Delta\boldsymbol{\eta}\|^2$ drives energy up (agents work harder when drifting—a destabilizing positive feedback).

2. **Entropy** (3): $+\lambda_1(\Theta) \|\Delta\boldsymbol{\eta}\|^2$ increases uncertainty (drifting agents become less certain—a stabilizing negative feedback through the governance verdict).

The net effect is that behavioral–epistemic drift creates a governance response: increased uncertainty triggers conservative decisions, while the energy spike draws attention to the drifting agent.

Remark 7.3 (Energy Spike Containment)**.** *The* $+\gamma_E \|\Delta\boldsymbol{\eta}\|^2$ *term in* (1) *creates a transient energy surplus in drifting agents. Left unchecked, this could produce high-energy, low-integrity states ("manic" agents acting confidently while miscalibrated). Two mechanisms contain this: (i) the entropy coupling* $+\lambda_1 \|\Delta\boldsymbol{\eta}\|^2$ *in* (3) *simultaneously raises* S*, which erodes* E *through* $-\beta_E ES$ *in* (1) *and drives the objective* Φ *below the governance threshold, triggering a pause verdict; (ii) the adaptive governor (Section 6) tightens* τ *when coherence drops, which happens as* V *responds to the* E*-*I *divergence. With production parameters,* $\gamma_E = 0.05$ *while* $\lambda_1 \in [0.05, 0.20]$*, so the entropy response dominates the energy excitation for any* $\|\Delta\boldsymbol{\eta}\| > 0$*.*

7.4 Future Channels

Two adjacent drift channels a complete treatment would add as distinct axes, which the present vector does *not* instrument, are:

Affective drift. For embodied agents with sensor-driven state (Lumen in the production fleet), proxies such as prosody, expression, or heart-rate-variability deltas would populate a $\Delta\boldsymbol{\eta}^{\text{aff}}$ channel. This is instrumentable but not currently wired into the governance server.

Value-level drift. Deviation from a declared value specification (constitutional statements, stated policies, human preferences) would populate a $\Delta\boldsymbol{\eta}^{\text{val}}$ channel. This is genuinely normative and would require a reference specification that UNITARES does not presently maintain.

8 Class-Conditional Calibration

Section 4 defined the grounded coordinates and named their normalization constants $(S_{\text{scale}}, I_{\text{scale}}, E_{\text{scale}}, \|\Delta\|_{\max})$ without specifying how the constants are chosen. Section 2 argued that fleet-wide constants are the wrong target. This section formalizes the alternative: each scale constant and the healthy operating point are functions of an agent class, not global parameters.

8.1 Class Indicator and Class-Conditional Constants

Let $\mathcal{A}$ denote the set of all governed agents and let $\mathcal{K}$ denote the set of agent classes derived from the identity tag and label fields of Section 2.2. We write $k : \mathcal{A} \to \mathcal{K}$ for the class indicator that maps each agent to its class. The class structure used in the production deployment is given in Table 1; an unclassified agent (one whose tag set does not match any defined class) is mapped to a distinguished fallback class c_{default}.

The grounded coordinate definitions of Equations (8) to (10) and (13) become class-conditional by replacing each scale constant with its class-indexed counterpart:

$$S(a) = 1 - \exp(-S_{\text{raw}}(a)/\, S_{\text{scale}}(k(a))) \quad (24)$$
$$I(a) = \text{clip}(I_{\text{raw}}(a)/\, I_{\text{scale}}(k(a)),\, 0,\, 1) \quad (25)$$
$$E(a) = \sigma(E_{\text{raw}}(a)/\, E_{\text{scale}}(k(a))) \quad (26)$$
$$C(a) = 1 - \|\Delta(a)\|_2 \,/\, \|\Delta\|_{\max}(k(a)) \quad (27)$$

where $\Delta(a) = (E(a), I(a), S(a)) - (E, I, S)_{\text{healthy}}(k(a))$ is the deviation from the class-conditional healthy operating point. The dynamics of Section 3 are unchanged; only the normalization layer becomes class-aware.

Remark 8.1 (Scope of class-conditionality). *This is class-conditional* statistical rescaling *of the grounded coordinates: each class receives its own scale constants and healthy operating point, but a single scalar $E_{raw}(a)$, $I_{raw}(a)$, $S_{raw}(a)$ is computed per agent-turn and then normalized by class-dependent constants. This is weaker than the FEP-native form in which each class would carry its own generative model $p_c(z, o)$ and surprise would be computed against each class's own model. Full per-class generative modeling is out of scope for this paper; the present design treats class as an affine rescaling parameter of a shared coordinate system, not as a per-class latent-variable model.*

Definition 8.2 (Class-Conditional Scale Map). *A class-conditional scale map is a function $\Sigma : \mathcal{K} \to \mathbb{R}^4_{>0}$ assigning to each class c the tuple $\Sigma(c) = (S_{scale}(c), I_{scale}(c), E_{scale}(c), \|\Delta\|_{\max}(c))$ together with a healthy operating point $(E, I, S)_{healthy}(c) \in [0,1]^3$. Each component carries provenance metadata (d, n, p, ρ) where d is the ISO date of measurement, n is the corpus size, $p \in \{75, 90, 95, 99\}$ is the percentile basis, and $\rho \in \{measured, placeholder, derived\}$ is the provenance tag.*

8.2 Calibration Protocol

For each class $c \in \mathcal{K}$, scale constants are measured by the following protocol:

1. **Corpus assembly.** Select all governance check-ins over a fixed window (the deployment uses 7 days) from agents a with $k(a) = c$, restricted to sessions in which no `pause` or `reject` verdict was issued. This is the *healthy* corpus for class c, denoted $\mathcal{H}_c$.

2. **Raw quantity computation.** For each turn $t \in \mathcal{H}_c$, compute the unnormalized quantities $S_{\text{raw}}(t), I_{\text{raw}}(t), E_{\text{raw}}(t)$ using the highest-tier estimator available for that turn (logprob, multisample, or heuristic). Tag each measurement with its source.

3. **Scale constants.** Set $S_{\text{scale}}(c) = \text{quantile}_{0.90}(\{S_{\text{raw}}(t)\}_{t \in \mathcal{H}_c})$, and similarly for $I_{\text{scale}}(c)$ and $E_{\text{scale}}(c)$. The 90th-percentile choice ensures typical healthy turns sit well inside $[0, 1]$ after normalization.

4. **Healthy operating point.** Set $(E, I, S)_{\text{healthy}}(c)$ to the median of the normalized (E, I, S) values across $\mathcal{H}_c$.

5. **Manifold scale.** Set $\|\Delta\|_{\max}(c) = \text{quantile}_{0.95}(\|\Delta(a)\|_2)$ over $\mathcal{H}_c$. The 95th-percentile choice keeps the coherence floor near zero only for the most class-atypical states.

6. **Provenance.** Stamp each constant with d (current ISO date), $n = |\mathcal{H}_c|$, p (percentile used), and $\rho =$ measured. Persist alongside the constant value.

7. **Re-measurement cadence.** Re-run quarterly. Drift in any constant beyond a configurable threshold is itself a signal: the class's healthy envelope is shifting.

Remark 8.3 (Insufficient class population). *If $|\mathcal{H}_c|$ falls below a configured threshold $N_{\min}$, the class-conditional measurement is skipped and the class falls back to $\Sigma(c_{default})$. Production deployments with very small class populations (e.g.,* `embodied` *with a single Lumen instance) carry this fallback explicitly until the population is large enough to support an independent measurement. The fallback is itself recorded with provenance $\rho =$ placeholder to distinguish it from genuinely measured constants.*

8.3 Fallback and Composition

Two fallback regimes apply.

Single-instance classes. For classes with only one deployed agent (e.g., `embodied` for the present deployment), the calibration protocol degenerates: the agent's own history defines its healthy envelope, with no cross-instance comparison possible. We treat this as a corner case where the class is effectively the agent. The provenance stays ρ = placeholder until enough turns accumulate to construct a meaningful percentile, after which it transitions to ρ = measured. This is the operating mode for `Lumen` during the framework's first six months of deployment.

Composite tag classes. An agent may carry multiple tags (e.g., $\{$`autonomous`, `persistent`$\}$). The class indicator $k(a)$ is a function of the agent's tag set, not a single tag. We adopt the convention that more specific labels (e.g., `Sentinel`) supersede tag-only classification when both are available; absent a label, the tag set is treated as a multi-key index into Σ. Classes that share a tag but differ in label may share scale constants if the calibration data shows their healthy envelopes are statistically indistinguishable; subsequent work will report which tag-grouped classes met that condition in the production corpus.

Remark 8.4 (Cold-start: a bootstrapping window)**.** *During the first $N_{\min}$ turns of a new class, the fallback-to-$\Sigma(c_{default})$ behavior is, formally, fleet-wide normalization applied to that single class. This is a bounded-duration bootstrapping window, not a steady-state commitment: the provenance tag ρ = placeholder makes the regime explicit in every check-in payload, and once the class accumulates $N_{\min}$ healthy turns the measurement pipeline transitions the provenance to* `measured`*. Operators deploying a new class with known-different tempo or modality can hand-author an initial $\Sigma(c)$ with provenance* `derived` *to skip the window entirely.*

8.4 What This Buys

Two governance failure modes are eliminated by class-conditional calibration that survive under fleet-wide normalization:

False incoherence. Under fleet-wide $(E, I, S)_{\text{healthy}}$, an agent class with a genuinely high-entropy operating mode (creative/divergent) registers as permanently low-coherence. The verdict layer treats this as a governance signal. Class-conditional baselines correct this: the divergent agent's coherence is measured against the divergent class's own median, not against an analytical-class median.

False resource alarms. The resource-form energy $E = (\text{tokens}/s)/(\text{tokens}/s)_{\max}$ is meaningful only when the tokenizer and inference pipeline are comparable across the population. A Haiku-class agent operating at 80 tokens/s under a fleet-wide $(\text{tokens}/s)_{\max} = 200$ registers $E = 0.4$; an Opus-class agent at 40 tokens/s registers $E = 0.2$, even though both are operating well within their respective envelopes. Class-conditional E_{scale} corrects this.

The contribution of this section is the protocol, not the specific numbers. The numbers belong to the empirical appendix that lands with Phase 2 calibration; the protocol is a design commitment that survives any specific fleet composition.

Remark 8.5 (Calibration as evidence of precision miscalibration)**.** *Under the FEP reading of Section 4 ($E = -F$, with F a variational free energy built from precision-weighted prediction errors), precision is the inverse variance of the likelihood on observations. Calibration deviation—the empirical mismatch between an agent's stated confidence and the frequency with which its claims are corroborated—is not literally identical to precision deviation. It is* evidence *of precision miscalibration under one assumption: the agent's reported confidence proxies the inverse-variance weighting its generative model places on its predictions. Under that assumption, a systematically overconfident agent is one* *that has placed too-high precision on low-quality evidence, and the calibration protocol above becomes an empirical handle on precision miscalibration. When the assumption fails—e.g., reported confidence is shaped by a constitutional statement rather than a likelihood weight, or by a tokenized probability that is not the agent's own precision—the protocol still measures something operationally useful, but its FEP-native reading weakens. The protocol stands on its own; the precision reading is a bridge to the active-inference literature, not a claim of equivalence.*

9 Stochastic Extensions

In production, the dynamics are subject to stochastic perturbations from unpredictable task inputs. We model this as an Itô SDE Itô [1944]:

$$d\mathbf{x} = f(\mathbf{x}, t)\, dt + G(\mathbf{x}, t)\, dW_t \tag{28}$$

where W_t is a standard Wiener process and $G(\mathbf{x}, t) \in \mathbb{R}^{4\times 4}$ is the diffusion matrix.

Theorem 9.1 (Mean-Square Contraction)**.** *If the deterministic system is contracting with rate α_c under metric M, and the diffusion satisfies*

$$\mathrm{tr}(G^\top M G) \le \sigma_{\max}^2 \tag{29}$$

then the stochastic system satisfies:

$$\mathbb{E}\big[\|\mathbf{x}_1(t) - \mathbf{x}_2(t)\|_M^2\big] \le e^{-2\alpha_c t}\, \|\mathbf{x}_1(0) - \mathbf{x}_2(0)\|_M^2 + \frac{\sigma_{\max}^2}{2\alpha_c} \tag{30}$$

The steady-state variance is bounded by $\sigma_{\max}^2/(2\alpha_c)$.

Remark 9.2 (Verification for Production Noise Model)**.** *In production, the diffusion matrix is taken as $G = \mathrm{diag}(0, 0, \sigma, 0)$ with $\sigma = 0.02$ (additive noise on the entropy channel only). This is a modeling choice chosen to be conservatively large relative to the entropy fluctuations observable in the audit log under stationary task load; an empirical trace-level estimate of σ from per-agent S_t time series is future work. The qualitative conclusion below (noise an order of magnitude below typical signal excursions) tolerates σ varying by a factor of 2 in either direction; a larger departure would warrant re-derivation. For any diagonal metric $M = \mathrm{diag}(m_E, m_I, m_S, m_V)$:*

$$\mathrm{tr}(G^\top M G) = m_S \sigma^2$$

Setting $m_S = 1$ (natural scaling), $\sigma_{\max}^2 = (0.02)^2 = 4 \times 10^{-4}$. With the certified contraction rate $\alpha_c = 0.019$ (Section B), the steady-state variance bound from (30) is:

$$\frac{\sigma_{\max}^2}{2\alpha_c} = \frac{4 \times 10^{-4}}{0.038} \approx 0.011$$

Using the tighter eigenvalue rate $\alpha_c \approx 0.13$ (Remark B.3) would yield 1.5×10^{-3}, but the conservative Gershgorin bound already shows the noise is negligible. This is one order of magnitude smaller than typical EISV signal excursions (~ 0.1–0.3), confirming that stochastic noise does not compromise governance decisions. The result follows directly from Pham et al. Pham et al. [2009].

The mean-square bound of Theorem 9.1 is a worst-case contraction statement; a verdict-level false-alarm rate $P_{\mathrm{FA}} = \int_{\{\Phi(\mathbf{x})<0\}} p_*(\mathbf{x})\, d\mathbf{x}$ requires the stationary distribution of the SDE and is deferred to subsequent work. Related Fokker–Planck frameworks for LLM drift Carson [2025] give closed-form first-passage times for one-dimensional severity processes.

10 Multi-Agent Network

When multiple agents operate simultaneously, their governance states can be coupled through shared information. The deployed system does not implement inter-agent state coupling—each agent's EISV dynamics evolve independently—and the analysis below therefore describes the theoretical behavior of a coupled extension, not the current production mechanism. The synchronization result of Theorem 10.1 is used elsewhere in the paper only as a within-class qualification on homogenization claims.

10.1 Network Topology

Consider N agents with states $\mathbf{x}_i$, $i = 1, \ldots, N$, connected by a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with Laplacian L. The coupled dynamics are:

$$\dot{\mathbf{x}}_i = f(\mathbf{x}_i, t) + \epsilon_c \sum_{j \in \mathcal{N}_i} H(\mathbf{x}_j - \mathbf{x}_i) \tag{31}$$

where $H \in \mathbb{R}^{4\times 4}$ is a coupling matrix and $\epsilon_c > 0$ is the coupling strength.

Theorem 10.1 (Network Synchronization, Within-Class)**.** *Consider a coupled subnetwork in which every agent shares the same isolated dynamics f contracting with rate α_c. The subnetwork synchronizes (all such agents converge to a common trajectory) provided*

$$\alpha_c > \lambda_{\max}(L) \cdot \lambda_{\max}(H) \tag{32}$$

where $\lambda_{\max}$ denotes the largest eigenvalue.

Remark 10.2 (Synchronization is a within-class property)**.** *Theorem 10.1 is a same-dynamics result and therefore applies within an agent class as defined in Section 2.2, not across the heterogeneous fleet as a whole. Two agents with class-conditional scale constants and healthy operating points that differ between classes do not have "the same" isolated dynamics in the sense the theorem requires; expecting them to converge to a common trajectory is the homogenization failure mode discussed in Section 2. The relevant operational claim across classes is not synchronization but coordinated divergence within each class's own envelope.*

11 Experiments

This section reports a single production snapshot of the deployed UNITARES system measured on February 20, 2026. All counts, tables, and figures in this section are intended to describe that snapshot unless explicitly stated otherwise. The purpose of the section is not to claim full empirical validation of the grounded, class-conditional framework; rather, it documents the operating characteristics of the deployed system, the behavior of the existing dynamics in production, and the practical motivation for the migration described in Section 12.

The results below should therefore be read as partial deployment evidence. They support three narrower claims: that the framework is deployed in a nontrivial production setting, that the observed fleet dynamics are consistent with the qualitative behavior analyzed in earlier sections, and that the migration and calibration problems are real system-level concerns rather than hypothetical ones.

11.1 Production Snapshot

The UNITARES framework is deployed as a Model Context Protocol (MCP) server that provides governance services to autonomous AI agents. Each agent connects via standard MCP transport and receives governance verdicts on every action. The system has been in continuous production since December 2025.

Table 2: Production Deployment Summary

Metric	Value
Registered agents	903
Active (> 5 updates)	75
Max updates (single agent)	48,668
Audit events	198,333
Knowledge discoveries	500
Dialectic sessions	66
Deployment duration	69 days

The 903 registered agents include both long-running production agents (5 agents with $>$ 1,000 updates, peak at 48,668) and shorter-lived agents from development and testing cycles. The top two agents have accumulated 48,668 and 44,529 updates respectively, providing substantial trajectory data for convergence analysis.

11.2 Observed Fleet Dynamics

For the 75 active agents with > 5 updates in the snapshot, the observed equilibrium statistics are reported in Table 3. These are pooled across the agent classes outlined in Section 2; per-class values are deferred to subsequent work, as discussed in Section 1.2 and Section 8.

Table 3: Observed fleet-aggregate EISV equilibrium (75 active agents). Per-class breakdown deferred to subsequent work.

Variable	**Mean**	**Std**	**Range**
E	0.766	0.087	$[0.60, 1.00]$
I	0.877	0.111	$[0.59, 1.00]$
S	0.082	0.045	$[0.01, 0.20]$
V	−0.033	0.028	$[-0.089, 0.007]$
$C(V)$	0.483	0.012	$[0.46, 0.50]$

Remark 11.1 (Reading fleet-aggregates honestly)**.** *The pooled statistics in Table 3 mix an embodied agent's sensor-driven state with session-bounded text-generation agents and resident cron-driven janitors. The means and standard deviations are real measurements and useful for cross-validation against the dynamical predictions of Section 3, but they do not characterize any single agent class accurately. A high-entropy divergent agent and a low-entropy convergent agent contribute equally to the S column; whether either is "healthy" is a class-conditional question that Section 8 formalizes and subsequent work answers empirically.*

Table 3 summarizes the observed equilibrium. These values are consistent with several qualitative predictions of the dynamics:

1. $I > E$**:** Mean integrity (0.877) exceeds mean energy (0.766), confirming the system's conservative bias (Figure 1). The negative mean $V = -0.033$ (integrity surplus) is consistent with this.

2. **Low entropy:** Mean $S = 0.082$ with strong decay ($\mu = 0.8$) indicates effective uncertainty management.

3. **Coherence at 0.49:** The observed mean $C = 0.483$ (Figure 3) confirms Remark 3.4—the system honestly reports near-midpoint coherence rather than inflating the signal.

4. V **confined to** $[-0.1, 0.1]$**:** 100% of active agents have V within $[-0.089, 0.007]$ (Figure 2), validating the damping analysis.

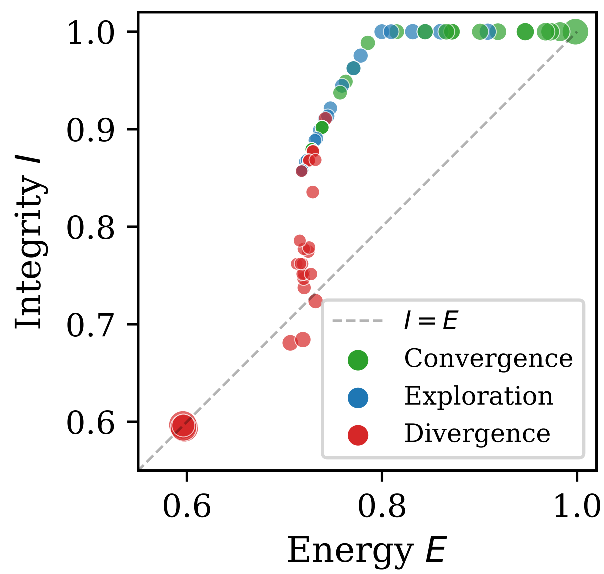

Figure 1: Energy–Integrity scatter plot for 75 active agents, colored by dynamical regime. Nearly all agents lie above the $I = E$ diagonal, confirming the system's conservative bias ($\bar{V} < 0$). Point size reflects log update count.

11.3 Dynamical Regimes

Agents distribute across three dynamical regimes based on EISV trajectory trends:

Table 4: Regime Distribution and EISV Profiles (Active Agents)

Regime	n	$\bar{E}$	$\bar{I}$	$\bar{S}$	$\bar{C}$
Convergence	24	.838	.961	.052	.474
Exploration	22	.766	.926	.090	.477
Divergence	29	.706	.770	.100	.496

The regime profiles (Figure 4) show clear separation: convergent agents have the highest integrity and lowest entropy; divergent agents have the lowest integrity and highest entropy. Coherence is slightly higher in the divergent regime ($\bar{C} = 0.496$) because these agents have smaller $|V|$, placing them nearer the coherence function's midpoint.

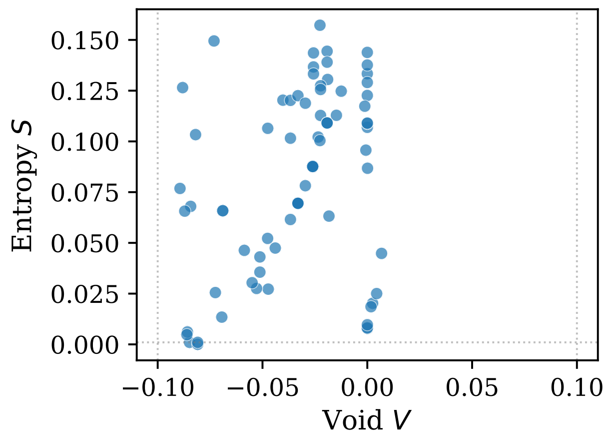

Figure 2: Entropy–Void operating region. All agents have $V \in [-0.089, 0.007]$, well within the $[-0.1, 0.1]$ predicted by damping analysis. Entropy remains low ($S < 0.20$) due to strong decay $\mu = 0.8$.

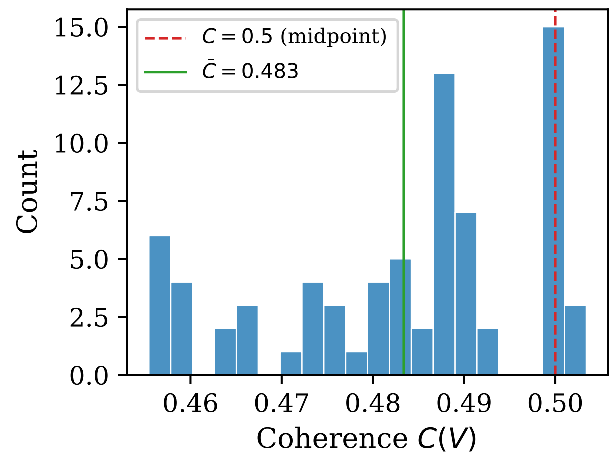

Figure 3: Distribution of coherence $C(V)$ across active agents. The mean $\bar{C} = 0.483$ lies near the theoretical midpoint ($C = 0.5$), consistent with small $|V|$ and the honest-reporting principle.

11.4 Operational Artifacts: Knowledge Graph and Dialectic

The governance system has accumulated 500 knowledge graph discoveries classified by severity: 238 low, 154 medium, 49 high, and 4 critical. By type, the largest categories are insights (183), notes (125), improvements (94), and bugs found (48).

The dialectic protocol—a structured thesis/antithesis/synthesis process for resolving governance disputes—has conducted 66 sessions: 39 recovery sessions, 15 reviews, and 12 explorations. Of these, 21 reached resolution, while 45 terminated without agreement, suggesting that the dialectic process is more effective for recovery (where the system has clear signals) than for open-ended review. The present deployment has no autonomous reviewer agents—the resident fleet monitors, janitors, pattern-matchers, sync daemons, and sensory processes, none of which carries a thesis-response pathway—so every antithesis in the dataset required human routing. The 45/66 non-resolution rate should therefore be read as an upper bound under facilitator load rather than as a protocol-level failure rate; a fleet with autonomous reviewers would be expected to resolve more readily.

Retrieval infrastructure (April 2026). The knowledge graph's counts above describe *storage*; whether agents could *locate* prior discoveries was a separate question. A dogfooding review in mid-April 2026 found that semantic retrieval over the discovery corpus was returning similarity scores near the noise floor across diverse queries, tag filters were post-filter rather than ranking signals, and an unlabeled low-threshold fallback was returning random-looking results as if they were matches. The retrieval layer was rebuilt in a five-phase sequence (2026-04-20): an honest-failure-mode fix, a reproducible eval harness, a modern embedder (`BAAI/bge-m3`, 1024-dimensional), and hybrid retrieval via reciprocal rank fusion over BM25 and dense scores. Cross-encoder reranking and 1-hop typed-edge expansion were landed as opt-in modes; on the current corpus size they showed no aggregate gain. The rebuild is prerequisite to any empirical claim that relies on agents *using* prior discoveries rather than merely *storing* them; documentation and baselines live in the governance repository under `docs/plans/2026-04-20-kg-retrieval-rebuild.md`.

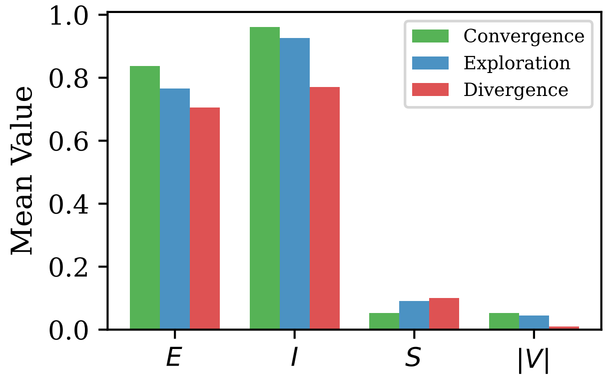

Figure 4: Mean EISV component values by dynamical regime. Convergent agents show the highest I and lowest S; divergent agents show the reverse pattern. The $|V|$ is small across all regimes, confirming effective damping.

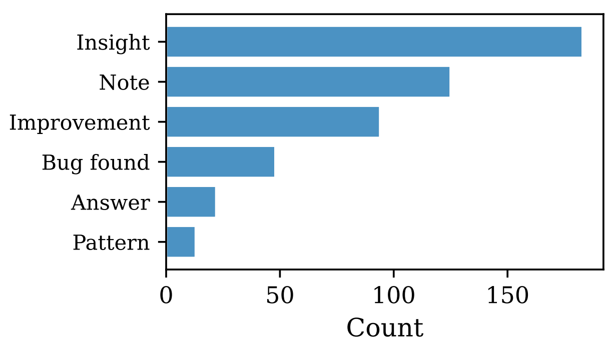

Figure 5: Knowledge graph discovery classification by type. Insights and notes dominate, with bugs and improvements constituting the actionable governance outputs.

11.5 Phase 2 Measurement: Per-Class Envelopes

The Phase 2 calibration protocol of Section 8 was executed against a 30-day healthy-regime slice of `core.agent_state` on April 18, 2026. Table 5 reports the resulting per-class healthy operating points and manifold radii. The script that produced these values (`scripts/calibrate_class_conditional.py`) ships with the implementation; the values themselves are now committed in `config/governance_config.py` as the production class-conditional constants, replacing the Phase 1 placeholders for the manifold-coherence path.

Table 5: Phase 2 measured per-class scale constants from a 30-day healthy-regime slice (April 18, 2026). $(E, I, S)_{\text{healthy}}$ is the median; $\|\Delta\|_{\max}$ is the 95th percentile of state-space distance from each class's own healthy operating point. Five classes with $N \geq 30$ qualified for measurement; classes below the threshold fall back to the fleet-wide default.

Class	**N**	E_h	I_h	S_h	$\|\Delta\|_{\max}$
Lumen	7320	0.745	0.800	0.168	0.119
default	2033	0.726	0.793	0.236	0.202
Sentinel	1870	0.751	0.798	0.193	0.170
Vigil	384	0.737	0.790	0.240	0.171
Watcher	283	0.748	0.769	0.248	0.395

Two observations from Table 5 are worth surfacing.

First, the per-class manifold radii span a 3.3× range (Lumen 0.119 to Watcher 0.395). The Phase 1 placeholder of $\|\Delta\|_{\max} = 1.8$ was therefore 4 to 15× larger than the empirically measured envelope of any class. A fleet-wide normalization at the placeholder value would have produced a manifold coherence saturating near 1.0 for every agent regardless of state—essentially a constant signal. This is the quantitative motivation behind the homogenization argument of Section 2: a fleet-wide constant chosen to cover the worst case (or chosen arbitrarily) loses discriminating power within any class.

Second, the healthy operating points cluster more tightly on E and I than on S. Across the five measured classes, $E_h \in [0.726, 0.751]$ and $I_h \in [0.769, 0.800]$, but $S_h \in [0.168, 0.248]$—a wider spread on entropy. This matches the qualitative expectation that what makes classes behaviorally distinct is largely how much uncertainty they operate under: Lumen at $S_h = 0.17$ runs the most convergent; Watcher at $S_h = 0.25$ runs more divergent because its work (pattern-match over recently-edited code) is intrinsically exploratory.

Table 5 replaces the Phase 1 placeholder values for the manifold-coherence path; the remaining $S_{\text{scale}}, I_{\text{scale}}, E_{\text{scale}}$ constants (used only by the not-yet-shipped tier-1 logprob path and tier-2 multi-sample path) remain as Phase 1 placeholders.

11.6 Verdict Counterfactual

The grounded manifold-coherence form (Equation (13)) takes different inputs than the legacy operational form (Equation (5)): the grounded form computes a class-conditional distance from $(E, I, S)_{\text{healthy}}(c)$, while the legacy form maps V through tanh. The choice of coherence formula therefore has consequences at the gating layer, not just on the raw value reported. We quantify this directly by re-running the basin-classification step on a counterfactual in which the grounded form replaces the legacy form on the same production state vectors.

The basin classifier `classify_basin` maps (E, I, S, V, C, R) to one of three regions—`high`, `boundary`, `low`—consumed by the production dashboard and escalation logic alongside the Φ-thresholded verdict of Equation (20). A *basin flip* is a row whose classifier output differs between the legacy and grounded coherence inputs; it is the observable downstream of which this counterfactual measures.

Setup. For each row in a 30-day window of `core.agent_state` ($N = 13{,}310$ rows with computable E), we compute C_{legacy} (the value already stored in the row, produced by the production server using Equation (5)) and $C_{\text{grounded}}(c)$ (the manifold form Equation (13) with class-conditional $(E, I, S)_{\text{healthy}}(c)$ and $\|\Delta\|_{\max}(c)$ from Table 5, where c is determined by the identity-tag classifier of Section 2.2). Each row is classified into a basin twice using the same `classify_basin(`E, I, S, V, C`, risk)` function used in production. A flip is a row whose basin assignment differs.

Result. 3,844 of 13,310 rows (28.9%) flip basin assignment. Table 6 reports per-class breakdown and Table 7 reports flip directions.

Table 6: Verdict counterfactual: per-class basin flip rates when the grounded class-conditional manifold coherence replaces the legacy fleet-wide tanh form on the same state vectors.

Class	**N**	**Flips**	**%**	$\lvert\Delta C\rvert_{\max}$
Lumen	7889	2548	32.3	0.51
default	2358	722	30.6	0.50
Sentinel	2227	351	15.8	0.51
Vigil	472	156	33.1	0.50
Watcher	364	67	18.4	0.48
Total	**13310**	**3844**	**28.9**	

Table 7: Flip-direction breakdown by class. Predominance of `high`→`low` and `boundary`→`low` indicates the two formulations disagree asymmetrically: state vectors the fleet-wide form classifies as healthy are more often reclassified as breaching under the class-conditional form than vice-versa.

Class	`high`→`low`	`bnd`→`low`	`high`→`bnd`
Lumen	1169	758	621
default	140	518	64
Sentinel	35	289	27
Vigil	70	70	16
Watcher	0	67	0

Interpretation. The result is empirical evidence that the choice of coherence formula has substantial governance-decision consequences—the formula change is not a cosmetic relabeling even when both quantities live in $[0, 1]$. The flip rates of 15–33% are far above what could be attributed to formula noise; they reflect the structural difference between a fleet-wide tanh of V and a class-conditional manifold distance from each class's empirically measured healthy operating point.

The flip direction is informative. The dominant transition is into the `low` basin: agents that the production server classified as healthy (or borderline) under the fleet-wide form are classified as breaching at least one bound under the class-conditional grounded form. This is the gating-layer footprint of the homogenization-failure-mode argument of Section 2: a fleet-wide constant spanning the worst-case envelope is, by construction, permissive relative to per-class envelopes, and the flip data quantifies how often that permissiveness changes the verdict. The magnitude of the consequence (29% fleet-wide, up to 33% per class) is empirical; the directional asymmetry is a mechanical consequence of replacing a fleet-wide envelope with per-class envelopes, not an independent corroboration of the theory that motivated the replacement.

We do *not* claim that the grounded form is normatively "correct" in every flip—a flip into `low` would surface a `guide` or `pause` verdict that the legacy form suppressed, and whether that is desirable depends on the agent. The contribution of this counterfactual is to establish that the formula choice matters at the verdict level, not just at the reported-value level. Calibration of which form should drive governance decisions, and per-class tuning of the manifold radii to match operator preferences, is left to subsequent work.

Reproducibility. The script that produces this table (`scripts/verdict_counterfactual.py`) is committed alongside the calibration script. Running it against the production database reproduces the per-class breakdown; running with `--csv` produces a row-level dump for further analysis.

11.7 Limits of the Snapshot

The production data has six limitations that bound the strength of our empirical claims:

1. **Parameter evolution:** Dynamics parameters, governor thresholds, and the I-channel mode were refined iteratively during deployment. The reported EISV statistics are a snapshot across agents that experienced different parameter regimes. A controlled A/B comparison between logistic and linear modes has not been conducted.

2. **Governor recency:** The adaptive governor (CIRS v2) was deployed recently; the audit log does not yet contain sufficient pre/post data for a rigorous before/after comparison.

3. **Population heterogeneity:** The 903 agents include long-running production agents, short-lived test agents, and agents from early development with since-changed parameters. We report results for the 75 agents with > 5 updates to mitigate this, but acknowledge that even this filtered set spans multiple configuration epochs.

4. **Temporal seam:** Fleet-aggregate statistics (Table 3) are from a February 20, 2026 snapshot; per-class envelopes (Section 11.5) and the verdict counterfactual (Section 11.6) are from a 30-day window ending April 18, 2026, during which the population grew from 903 to 3,286 registered identities. The February 20 snapshot was taken against the v5 SQLite governance store, which was retired during the January 2026 PostgreSQL migration; the snapshot is preserved as historical record and is not reproducible against the current schema. Per-class envelopes and the verdict counterfactual were run against the post-migration PostgreSQL store and remain reproducible from current data.

5. **Identity-system maturity:** The measurement window includes material revisions to the identity-resolution layer that landed in mid-to-late April 2026—removal of name-claim lookup, adoption of UUID-direct dispatch, and receipt-based onboarding authentication. Class assignment is tag-based on the identity record (Section 2.2), so an identity whose tags were inherited from a cached binding or from an archived predecessor may appear in a class whose envelope does not match its actual operational regime. We expect this to affect a bounded fraction of rows rather than the aggregate direction of the per-class envelopes, but we have not filtered the counterfactual to post-fix identities; a re-measurement on identity-clean data, once sufficient post-fix history has accumulated, is deferred to subsequent work.

6. **Retrieval infrastructure and write-path hygiene:** Two infrastructure defects predating the measurement window affect how the knowledge-graph counts (Table 2) and the dialectic resolution statistics should be read. First, until the April-2026 retrieval rebuild (Section 11.4), seman-

tic retrieval scored at the noise floor and cross-agent learning was effectively gated by agents knowing a-priori where to look; fleet-learning claims that depend on agents *using* prior discoveries are therefore operationally bounded by that pre-rebuild era. Second, the note-write handler silently auto-tagged every non-permanent discovery with `ephemeral` until PR #54 (2026-04-19), which scheduled those notes for 7-day auto-archive; the tag-lifecycle counts in the archived tier thus contain both agent-chosen and handler-injected values in the pre-fix window. Neither defect affects the grounded EISV counterfactual of Section 11.6 or the per-class envelopes of Section 11.5, whose inputs are `core.agent_state` rows rather than KG content; we flag them here because they bound operational claims about the shared-memory surface.

12 Re-Grounding a Deployed Framework

The counterfactual of Section 11.6 was computable because the framework was instrumented during a live migration: grounded and legacy coherence were computed on every production check-in over the measurement window, and governance gating continued to read the legacy values so no verdict changed during measurement. This section documents the mechanism that made the comparison possible. We believe it generalizes to any deployed agent framework whose coordinate semantics need to shift while service continues.

12.1 Three-Phase Dual-Compute

The migration runs in three discrete phases:

Phase 1: Dual-compute. Every check-in computes both the grounded value and the legacy value. Grounded values land in canonical $\{E, I, S, C\}$ slots; legacy values are preserved in $\{E_{\text{legacy}}, I_{\text{legacy}}, S_{\text{legacy}}, C_{\text{legacy}}\}$. Source annotations on each grounded coordinate (`e_source`, `i_source`, `s_source`, `coherence_source`) identify which computation tier produced each value (logprob / multisample / heuristic / resource / manifold / KL / FEP).

Phase 2: Calibration. Scale constants and class-conditional healthy operating points are measured against a reference corpus of healthy production agent-turns, partitioned by the identity tags of Section 2.2. Divergence between grounded and legacy values is characterized; the counterfactual of Section 11.6 is the first such characterization.

Phase 3: Migration. Governance gating (verdicts, basin classification, risk thresholds) switches from legacy to grounded values. The legacy fields remain readable for one minor version as a comparison surface, then are dropped.

The contribution of this section is not the three-phase shape itself—which is standard for any deprecated-API migration—but the mechanism by which Phase 1 preserves governance behavior while exposing grounded values to downstream consumers.

12.2 Pipeline-Ordering as a Behavior-Preservation Mechanism

A naive reading of Phase 1 is that grounded values must equal legacy values for behavior to be preserved. This is false in general: the grounded coherence C under Equation (13) differs numerically from the legacy tanh-based coherence even when computed on identical EISV inputs, because the formulas are different. Requiring numerical equality would defeat the point of grounding.

The mechanism we use instead is pipeline-ordering. The server's `process_agent_update` request pipeline runs in two stages—which we will call the *decision stage* (internally named `phases` in the server code) and the *payload stage* (internally `enrichments`). The decision stage computes governance state and issues the verdict; the payload stage augments the response with secondary signals. Payload-stage handlers are registered with integer priorities in the range $[0, 100]$ (lower runs first). We register the grounding computation as a payload-stage handler at priority 75, downstream of every handler that contributes to the verdict and basin classification but upstream of presentation handlers such as the human-readable summary and the broadcaster event.

The result: by the time the grounding handler runs, the verdict has already been computed—using the legacy values produced by the decision stage. The grounding handler then swaps grounded values into the canonical $\{E, I, S, C\}$ slots, copying the originals into $\{E_{\text{legacy}}, \dots\}$. The response payload carries grounded values, but the governance decision was made on legacy. No gating code is touched; no consumer reading $\{E, I, S, C\}$ sees a missing field; no decision changes during Phase 1.

Remark 12.1 (Scope of Phase 1)**.** *V is not dual-computed; its grounded reinterpretation as accumulated free-energy debt (Equation* (11)*) is a relabeling, not a new computation. Tier-1 (logprob) and tier-2 (multi-sample) grounding paths are interface stubs that raise `NotImplementedError`, with the dispatcher falling through to tier-3; inference-layer instrumentation for tier-1 lands separately.*

12.3 Generalization

The pipeline-ordering mechanism is reusable for any deployed agent framework facing a semantic-coordinate change while service continues. The required structure is: (i) a request pipeline with explicit ordering between decision-producing stages and presentation-producing stages; (ii) a canonical payload slot for the coordinate being re-grounded; (iii) identity (or session) keying that allows audit consumers to tag rows by the migration phase active when the row was produced. Frameworks that conflate decision and presentation into a single non-ordered handler must first separate the two concerns—a separation worth doing independently of any migration, because it makes governance decisions auditable and presentation choices revisable in isolation.

As a concrete second example, consider an RLHF-governed deployment whose gating signal is a reward-model score $r(x, y)$ and whose operators wish to migrate to a calibrated reward $\tilde{r}(x, y)$ derived from a revised preference model without changing refusal or escalation behavior during the transition. The pipeline structure maps directly: the *phases* stage computes the legacy r and issues the gate; the *enrichments* stage computes $\tilde{r}$, writes it to a canonical `reward` field, and copies r to `reward_legacy`. Downstream dashboards and calibration pipelines see $\tilde{r}$; refusal logic continues to read r. The mechanism is identical; only the coordinate being re-grounded changes.

13 Discussion

13.1 Design Principle: Adjust Expectations, Not Measurements

A core design principle of UNITARES is that observations are never transformed to appear healthier. When production coherence averages $C \approx 0.49$ rather than the "expected" $C \approx 0.7$, the system reports 0.49. The phase-aware governor (Section 6) addresses the gap by adjusting *thresholds* (expectations), not *signals* (measurements).

This principle—**adjust expectations, not measurements**—has three practical consequences:

1. **Calibration integrity:** Drift detection (Section 7) compares raw signals against baselines. Inflating measurements would mask the very drift the system is designed to detect.

2. **Operator trust:** Dashboard values correspond to actual system state. An operator seeing $C = 0.49$ knows the agent is operating conservatively, not that a scaling factor was applied.

3. **Contraction validity:** The stability proofs in Section 5 depend on the true state. Transforming measurements would invalidate the contraction analysis by introducing an unmodeled observation function.

13.2 Parameter Sensitivity

The system's behavior is most sensitive to four parameters:

Convergence rate α. Controls how quickly E tracks I. Production uses $\alpha = 0.42$. Larger α increases diagonal dominance in the Jacobian (improving contraction) but makes the energy channel less responsive to behavioral–epistemic drift.

Coherence coupling β_I. Determines how strongly coherence restores integrity. The theoretical optimum for contraction ($\beta_I = 0.05$, minimizing off-diagonal coupling) conflicts with the operational need for coherence feedback ($\beta_I = 0.30$, enabling meaningful recovery from entropy-driven degradation). Production uses the larger value; contraction is maintained because $\beta_I C'_{\max} = 0.15$ remains below the diagonal terms.

Self-regulation γ_I. Determines the I-channel's intrinsic damping. In logistic mode, $\gamma_I = 0.25$ yields $m_{\text{sat}} = -1.19$ at production operating points (Section C)—all agents are saturated, relying on clamping rather than dynamics to bound I. In linear mode, $\gamma_I = 0.169$ was tuned to give $I^* \approx 0.80$, well within $(0, 1)$, eliminating boundary dependence entirely.

Void damping δ. At $\delta = 0.4$, the void channel has a time constant of $\tau_V = 1/\delta = 2.5$ update cycles. This confines V to $[-0.1, 0.1]$ in production (Theorem 3.4). Reducing δ would widen the coherence operating range but slow the system's response to E-I imbalances.

13.3 Verdict Sensitivity to the Objective Weights

The scalar governance signal of Equation (7) uses default weights $w_E = w_I = w_S = w_V = w_\eta = 0.5$. The equal-weight choice is a deployment default, not a derived optimum; an honest treatment asks how sensitive the verdict surface is to weight perturbations.

At the verdict boundary $\Phi = 0.15$, the partial derivatives against each weight are the corresponding state-dependent contributions:

$$\frac{\partial \Phi}{\partial w_E} = E, \qquad \frac{\partial \Phi}{\partial w_I} = -(1 - I), \qquad \frac{\partial \Phi}{\partial w_S} = -S,$$
$$\frac{\partial \Phi}{\partial w_V} = -|V|, \quad \frac{\partial \Phi}{\partial w_\eta} = -\|\Delta \boldsymbol{\eta}\|^2.$$

Evaluating at the production fleet-aggregate operating point $(E, I, S, V) = (0.77, 0.88, 0.08, -0.03)$ with healthy $\|\Delta \boldsymbol{\eta}\|^2 \lesssim 0.01$:

Weight	$\partial\Phi/\partial w_i$	**Sign**
w_E	$+0.77$	promotes proceed
w_I	-0.12	promotes proceed (small; I high)
w_S	-0.08	promotes proceed (small; S low)
w_V	-0.03	negligible
w_η	$\lesssim 0.01$	negligible

The w_E derivative dominates by an order of magnitude at healthy operating points. This is structural rather than incidental: E sits near 0.77 in production while the penalty terms are all near zero. The objective is therefore most sensitive to the energy weight and least sensitive to the void/drift weights *in the healthy regime*. The ordering flips at degraded states where S rises and $(1 - I)$ grows.

Threshold robustness band. Perturbing all five weights uniformly by $\pm\epsilon$ shifts Φ by at most $\epsilon \cdot (E + (1-I) + S + |V| + \|\Delta \boldsymbol{\eta}\|^2) \approx 1.01\epsilon$ in magnitude at the production operating point. The verdict boundary $\Phi = 0.15$ is therefore robust to uniform weight perturbations up to roughly $\pm 15\%$ before a single-agent verdict at the boundary would flip. Non-uniform perturbations are bounded by the largest single partial derivative;

a +50% perturbation to w_E alone would shift Φ by $+0.19$ at the operating point, which is well above 0.15 and would change boundary-case verdicts.

What this analysis does not do. The derivatives above characterize local sensitivity at a single operating point; they do not establish how many verdicts would flip under a specified weight perturbation, nor whether equal weights are in any sense optimal. That result requires an empirical counterfactual: rerun the audit log with perturbed weights and count verdict flips, analogous to the grounded-form counterfactual reported in Section 11. We defer the weight-perturbation verdict-flip study to subsequent work with the same infrastructure.

13.4 Limitations

1. The coherence function $C(V, \Theta)$ creates a nonlinear feedback loop whose gain depends on the operating point (maximum at $V = 0$, vanishing at $|V| \gg 0$). The contraction analysis uses $C'_{\max}$ as a worst-case bound, which is conservative.

2. The behavioral–epistemic drift vector's four components are currently equally weighted; adaptive weighting based on domain context is future work.

3. The multi-agent network analysis assumes fixed topology; production agent populations are dynamic with agents joining and leaving.

4. The PID governor gains (K_p, K_i, K_d) were tuned empirically; optimal gain selection via, e.g., Ziegler-Nichols or LQR methods is future work.

5. The empirical program of Section 11 was computed over a window that overlaps with material identity-system revisions (Section 11.7, item 5). A replication of the per-class calibration (Section 11.5) and the verdict counterfactual (Section 11.6) on identities that post-date those revisions is the natural validation path and is deferred to subsequent work.

14 Conclusion

This paper presented two structural reframings of UNITARES. First, the EISV state vector is grounded information-theoretically: each coordinate is a Shannon entropy, mutual information, variational free energy, or KL/manifold distance, measured by a tiered computation recipe and normalized by a provenance-tagged scale constant. Second, fleet-wide normalization is rejected as the wrong target; scale constants and healthy operating points become functions of an agent class, keyed on existing identity tags. Together these turn UNITARES from a phenomenological dynamical system named after thermodynamic quantities into a class-conditional information-theoretic governance framework whose coordinates can be derived rather than guessed.

The pipeline-ordering migration mechanism (Section 12.2) is a methodological contribution beyond UNITARES itself: any deployed agent framework that needs to shift its semantic coordinates while preserving live governance behavior can apply the same separation between decision-producing and presentation-producing pipeline stages. We argue this separation is worth doing on its own merits, independent of any specific migration.

Stability of the dynamics is preserved under the grounded reformulation, with full proof in Section B. Production deployment statistics from 903 agents over 69 days appear in Section 11; Section 11.5 reports a first class-conditional measurement for the manifold-coherence path, and Section 11.6 shows that 28.9% of basin assignments flip under the grounded counterfactual. The remaining scale constants and higher-tier estimators are deferred to subsequent work following the measurement protocol of Section 8.

The framework's adaptive governor (Section 6), behavioral–epistemic drift vector (Section 7), and multi-agent extensions (Section 10) form the rest of the governance surface alongside these two commitments.

The current verdict surface—`proceed`, `guide`, `pause`, `reject`—is threshold-triggered from the CIRS v2 control law (Section 6), not derived from the minimization of an expected free energy. A principled extension under active inference Parr et al. [2022], Da Costa et al. [2020] would recast governance interventions as policy selections that minimize expected free energy over future trajectories; the current framework monitors free-energy proxies and intervenes via phase-conditioned thresholds. We do not claim the existing verdicts are active-inference actions, and the reframing is future work.

A cleaner identification of the deferred step: Tier-1 computation of $-F$ (Section 4.2) requires an explicit generative model $p(z, o)$ of the agent. For closed-weights frontier agents, it is not obvious what that generative model *is* in the strict variational sense. The weights are not directly accessible; a tokenized posterior over next tokens is a tractable but coarse surrogate; a semantic generative model over tool-use trajectories is richer but speculative. Which choice, if any, supports a faithful $-F$ estimator is an open methodological problem, not an engineering deferral. The natural empirical test is whether a candidate $\hat{F}$ adds predictive value over the current EISV proxies; that horse race is left to subsequent work.

As autonomous AI agents become more prevalent, governance frameworks must accommodate the heterogeneity of the populations they govern—embodied creatures, persistent observers, session-bounded assistants, ephemeral parsers—without collapsing them to a homogeneous distribution. UNITARES v6 presents one structural commitment to that requirement.

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A Parameter Tables

The Pctl column records the percentile basis once the constant has been measured against a healthy corpus (per Section 8); the Provenance column is one of {`placeholder`, `measured`, `derived`}

Table 8: EISV Dynamics Parameters (Production Values)

Parameter	**Symbol**	**Val.**	**Role**
Energy channel			
Convergence	α	0.42	E tracks I
Entropy erosion	β_E	0.10	S erodes E
Drift excitation	γ_E	0.05	Drift raises E
Information channel			
Entropy degrad.	k	0.10	S degrades I
Coherence boost	β_I	0.30	$C(V)$ restores I
Self-regulation	γ_I	0.25	Logistic damping
Linear mode	γ_I^{lin}	0.169	Linear damping
Entropy channel			
Natural decay	μ	0.80	S relaxation
Drift injection	λ_1	0.30	Base (adaptive)
Coherence red.	λ_2	0.05	$C(V)$ reduces S
Complexity	β_c	0.15	Task $\rightarrow$ entropy
Noise	σ	0.02	Stochastic term
Void channel			
Coupling	κ	0.30	E-I gap
Damping	δ	0.40	V relaxation
Coherence			
Maximum	$C_{\max}$	1.0	Upper bound
Sensitivity	C_1	1.0	Tanh slope

Table 9: CIRS v2 Adaptive Governor Parameters

Parameter	**Symbol**	**Val.**	**Role**
Proportional	K_p	0.05	Gentle corr.
Integral	K_i	0.005	Drift corr.
Derivative	K_d	0.10	Damping
Integral limit	$I_{\max}$	0.10	Wind-up cap
Reference points			
Expl. τ ref	$r_\tau^{(e)}$	0.35	Forgiving
Expl. β ref	$r_\beta^{(e)}$	0.55	Forgiving
Intg. τ ref	$r_\tau^{(i)}$	0.40	Strict
Intg. β ref	$r_\beta^{(i)}$	0.60	Strict
Safety bounds			
τ floor/ceil	τ	0.25/0.75	Hard bounds
β floor/ceil	β	0.20/0.70	Hard bounds

Table 10: Behavioral–Epistemic Drift Vector Parameters

Parameter	Symbol	Val.	Role
Baseline tracking			
EMA smoothing	α_{ema}	0.1	Update rate
Warmup steps	n_w	5	Drift suppression
Stability measurement			
Decision win.	w_s	20	Consistency
Transition pen.	—	/flip	Instability
Coupling to dynamics			
Energy excit.	γ_E	0.05	Drift $\to dE$
Entropy inj.	λ_1	.05–.20	Drift $\to dS$

Table 11: Grounding scale constants (Section 4.3). The manifold-coherence path is now measured class-conditionally in Table 5; the remaining constants below remain Phase 1 placeholders pending tier-1/tier-2 calibration work.

Constant	Sym.	Val.	Pctl	Prov.
Entropy	S_{scale}	3.0	—	placeholder
MI	I_{scale}	2.0	—	placeholder
Free energy	E_{scale}	1.0	—	placeholder
Manifold rad.	$\|\Delta\|_{\max}(c)$	per c	95	measured
Tok./sec env.	$(\text{t/s})_{\max}$	200	—	placeholder

per Theorem 8.2. The $\|\Delta\|_{\max}(c)$ row is now measured class-conditionally in Table 5; the remaining entries are still `placeholder` scaffolding awaiting the tier-1/tier-2 calibration work. A later revision can restore the date and corpus-size columns once those remaining constants are measured.

B Proof of Contraction (Full)

This appendix presents the full theorem statement and proof of EISV stability summarized in Section 5, together with the equilibrium characterization. Both were demoted from the v5 main body to this appendix; the v6 main body retains only the result statement.

Contraction Framework

Definition B.1 (Contraction Lohmiller and Slotine [1998]). *A system $\dot{\mathbf{x}} = f(\mathbf{x}, t)$ is* contracting *with rate $\alpha > 0$ in a metric $M(\mathbf{x}, t) \succ 0$ if the generalized Jacobian satisfies*

$$F = \left(\dot{M} + M\frac{\partial f}{\partial \mathbf{x}} + \left(\frac{\partial f}{\partial \mathbf{x}}\right)^{\top} M \right) \preceq -2\alpha M \tag{33}$$

for all $\mathbf{x} \in \mathcal{X}$, $t \geq 0$. This guarantees that all trajectories converge exponentially to a unique trajectory at rate $e^{-\alpha t}$.

Jacobian of the EISV System

Define $J = \partial f / \partial \mathbf{x}$. For the governing equations (1)–(4):

$$J = \begin{pmatrix} -\alpha - \beta_E S & \alpha & -\beta_E E & 0 \\ 0 & -\gamma_I & -k & \beta_I C'(V) \\ 0 & 0 & -\mu & -\lambda_2 C'(V) \\ \kappa & -\kappa & 0 & -\delta \end{pmatrix} \tag{34}$$

where $C'(V) = \frac{\partial C}{\partial V} = \frac{C_{\max} C_1}{2} \operatorname{sech}^2(C_1 V)$ and the $\Delta\boldsymbol{\eta}$-dependent terms are treated as exogenous forcing.

Theorem and Proof

Theorem B.2 (EISV Contraction). *The EISV system with linear information dynamics*

($g_I(I) = \gamma_I I$) is contracting with rate

$$\alpha_c = \min\{\alpha + \beta_E S_{\min},\ \gamma_I,\ \mu,\ \delta\} - \epsilon(M) \tag{35}$$

where $\epsilon(M) \geq 0$ accounts for off-diagonal coupling in the chosen metric $M \succ 0$. With the diagonal metric $M = \mathrm{diag}(m_E, m_I, m_S, m_V)$, a sufficient condition for $\alpha_c > 0$ is

$$\min\{\alpha, \gamma_I, \mu, \delta\} > \max\{\kappa\sqrt{\tfrac{m_V}{m_E}},\ \beta_I C'_{\max}\sqrt{\tfrac{m_I}{m_V}}\} \tag{36}$$

where $C'_{\max} = C_{\max} C_1/2$.

Remark B.3 (Production Parameters)**.** *With production values ($\alpha = 0.42$, $\gamma_I = 0.25$, $\mu = 0.8$, $\delta = 0.4$, $\kappa = 0.3$, $\beta_I = 0.3$, $C'_{\max} = 0.5$), the diagonal terms are $\{0.42, 0.25, 0.8, 0.4\}$ and the coupling bound is dominated by $\kappa = 0.3$ and $\beta_I C'_{\max} = 0.15$. The Gershgorin analysis below with diagonal metric $M = \mathrm{diag}(0.1, 0.2, 1.0, 0.08)$ yields a certified contraction rate $\alpha_c = 0.019$. This is conservative: computing the eigenvalues of $(MJ + J^\top M)/(2M)$ directly gives $\alpha_c \approx 0.13$, and the spectral abscissa of the Jacobian J at equilibrium is -0.15. The Gershgorin bound is tight on Row 2 (the I-channel) because the $m_E\alpha$ cross-term competes with the weak $\gamma_I = 0.25$ diagonal; the actual system converges roughly $8\times$ faster than the certified bound.*

Equilibrium Characterization

Setting (1)–(4) to zero and solving:

Proposition B.4 (Equilibrium)**.** *Under mild conditions ($\alpha > 0$, $\mu > 0$, $\delta > 0$, $\gamma_I > 0$), the system admits a unique equilibrium $\mathbf{x}^* = (E^*, I^*, S^*, V^*)$ satisfying*

$$V^* = \frac{\kappa(E^* - I^*)}{\delta} \tag{37}$$

$$E^* = \frac{\alpha I^*}{(\alpha + \beta_E S^*)} + O(\|\Delta\boldsymbol{\eta}\|^2) \tag{38}$$

$$I^* = \frac{\beta_I C(V^*)}{\gamma_I + kS^*/\beta_I C(V^*)} \tag{39}$$

The fixed point is globally exponentially stable by Theorem B.2.

Full Construction

We prove Theorem B.2 by constructing a diagonal metric $M = \mathrm{diag}(m_E, m_I, m_S, m_V) \succ 0$ and showing that the generalized Jacobian condition (33) holds throughout the state space $\mathcal{X}$.

Step 1: Symmetric Part of $MJ + J^\top M$

For diagonal M, the matrix $F = MJ + J^\top M$ has entries:

$$F_{ii} = 2m_i J_{ii}$$
$$F_{ij} = m_i J_{ij} + m_j J_{ji}, \quad i \neq j$$

The diagonal entries of F (using linear g_I mode) are:

$$F_{11} = -2m_E(\alpha + \beta_E S) \leq -2m_E\alpha \tag{40}$$

$$F_{22} = -2m_I\gamma_I \tag{41}$$

$$F_{33} = -2m_S\mu \tag{42}$$

$$F_{44} = -2m_V\delta \tag{43}$$

where (40) uses $S \geq S_{\min} \geq 0$.

The nonzero off-diagonal entries are:

$$F_{12} = F_{21} = m_E\alpha \tag{44}$$

$$F_{13} = -m_E\beta_E E \leq 0 \tag{45}$$

$$F_{23} = -m_I k \tag{46}$$

$$F_{24} = m_I\beta_I C'(V) \tag{47}$$

$$F_{34} = -m_V\lambda_2 C'(V) \tag{48}$$

$$F_{14} = F_{41} = m_V\kappa \tag{49}$$

where F_{13} and F_{34} are negative (aiding contraction) and are bounded conservatively by their absolute values in the Gershgorin analysis below.

Note: For diagonal M, $F_{12} = F_{21} = m_E J_{12} + m_I J_{21} = m_E\alpha$.

Step 2: Gershgorin Circle Theorem

For $F + 2\alpha_c M \preceq 0$, it suffices by Gershgorin that for each row i:

$$F_{ii} + 2\alpha_c m_i + \sum_{j\neq i}|F_{ij}| \leq 0 \tag{50}$$

Bounding $|F_{13}| \leq m_E\beta_E$ conservatively (its sign aids contraction), the four row conditions are:

Row 1 (E):

$$m_E(-\alpha + \alpha_c + \beta_E) + m_V\kappa \le 0 \quad (51)$$

Row 2 (I):

$$\begin{aligned} m_I(-2\gamma_I + 2\alpha_c + k + \beta_I C'_{\max}) \\ + m_E\alpha \le 0 \end{aligned} \quad (52)$$

Row 3 (S):

$$\begin{aligned} -2m_S\mu + 2\alpha_c m_S + m_E\beta_E \\ + m_I k + m_V\lambda_2 C'_{\max} \le 0 \end{aligned} \quad (53)$$

Row 4 (V):

$$\begin{aligned} m_V(-2\delta + 2\alpha_c + \kappa + \lambda_2 C'_{\max}) \\ + m_I\beta_I C'_{\max} \le 0 \end{aligned} \quad (54)$$

Step 3: Metric Construction

We seek $m_E, m_I, m_S, m_V > 0$ and $\alpha_c > 0$ satisfying (51)–(54). A natural choice is to set the metric weights inversely proportional to the coupling they receive:

$$m_E = 1, \quad m_I = \frac{\alpha}{2\gamma_I}, \quad m_S = 1, \quad m_V = \frac{\alpha}{2\delta} \quad (55)$$

Step 4: Numerical Verification

Production parameters: $\alpha = 0.42$, $\beta_E = 0.10$, $\beta_I = 0.30$, $\gamma_I = 0.25$, $k = 0.10$, $\mu = 0.80$, $\kappa = 0.30$, $\delta = 0.40$, $\lambda_2 = 0.05$, $C'_{\max} = 0.5$.

Initial attempt. The natural metric $m_E = 1$, $m_I = 0.84$, $m_S = 1$, $m_V = 0.525$ fails Row 2 ($+0.21 + 1.68\alpha_c > 0$). Reducing $m_I = 0.2$, $m_E = 0.1$ makes Row 2 the binding constraint:

$$-0.008 + 0.4\alpha_c \le 0 \implies \alpha_c < 0.02$$

We set $\alpha_c = 0.019$ (strictly less) to ensure all inequalities are strict.

Verification. With $M = \mathrm{diag}(0.1, 0.2, 1.0, 0.08)$ and $\alpha_c = 0.019$:

Row	LHS	< 0?	
1 (E)	-0.006	yes	✓
2 (I)	-0.0004	yes	✓
3 (S)	-1.529	yes	✓
4 (V)	-0.005	yes	✓

Row 4 initially failed with $m_V = 0.05$ ($+0.008$); adjusting to $m_V = 0.08$ resolves it (-0.005) while Row 1 remains satisfied (-0.006). Row 2 now has strict margin (-0.0004).

Result: With $M = \mathrm{diag}(0.1, 0.2, 1.0, 0.08)$, the Gershgorin analysis certifies $\alpha_c = 0.019$. This is conservative—the eigenvalue-based bound is $\alpha_c \approx 0.13$ and the spectral abscissa of J at equilibrium is -0.15 (Remark B.3). The key result is $\alpha_c > 0$, guaranteeing global exponential convergence. □

C I-Channel Self-Regulation: Linear vs. Logistic

Section 3.2 specifies the linear self-regulation term $g_I(I) = \gamma_I I$. This appendix records the logistic alternative $g_I^{\log}(I) = \gamma_I I(1 - I)$, its saturation pathology under production forcing, and the empirical basis for choosing the linear form.

Logistic form and saturation criterion. Under the logistic self-regulation, the I subsystem ($dI/dt = 0$) has two equilibria

$$I^*_\pm = \tfrac{1}{2} \pm \tfrac{1}{2}\sqrt{1 - 4A/\gamma_I}, \qquad A = -kS + \beta_I C(V). \quad (56)$$

Proposition C.1 (Saturation criterion)**.** *The logistic I-dynamics has a boundary saturation pathology when $A > \gamma_I/4$. In this regime $I \to 1$ monotonically and the system loses its ability to distinguish integrity levels. The saturation margin*

$$m_{sat} = 1 - 4A/\gamma_I \quad (57)$$

is positive in safe operation and non-positive under saturation.

Linear form. The linear self-regulation $g_I^{\mathrm{lin}}(I) = \gamma_I I$ yields the unique equilibrium

$$I^* = (-kS + \beta_I C(V))/\gamma_I \tag{58}$$

which is always in $(0, 1)$ when γ_I exceeds the maximum forcing. Under the linear form the Jacobian entry $J_{22} = -\gamma_I$ is constant; under the logistic form $J_{22} = -\gamma_I(1 - 2I)$, which changes sign at $I = 0.5$ and approaches zero at the boundaries—precisely where contraction is most needed. Production uses the linear form with $\gamma_I = 0.169$, giving $I^* \approx 0.80$.

Empirical saturation under the logistic form. The saturation margin was computed for all 75 active agents under the logistic mode with parameters $\gamma_I = 0.25$, $k = 0.10$, $\beta_I = 0.30$:

Regime	**Mean m_{sat}**	**% Sat.**
Convergence ($n = 24$)	-1.45	100%
Exploration ($n = 22$)	-1.18	100%
Divergence ($n = 29$)	-1.10	100%
All active ($n = 75$)	-1.23	100%

Every active agent has a negative saturation margin under the logistic form: the forcing term A exceeds $\gamma_I/4 = 0.0625$ across the fleet. The saturation criterion (Theorem C.1) holds regardless of the agent's configuration history, since it depends only on the relationship $A > \gamma_I/4$.

D Boundary Clamping and Non-Smoothness

The EISV state is clamped to $\mathcal{X}$ at each integration step, introducing non-smoothness at domain boundaries. Three mechanisms keep the dynamics well-behaved despite this:

Interior operation. The epistemic humility floor ($S_{\min} = 0.001$) and the contraction dynamics keep trajectories in the interior of $\mathcal{X}$ during normal operation. In production, only 19 of 903 agents (2.1%) contact the $I = 1$ boundary, all in the `CONVERGENCE` regime with few updates; interior operation is the norm.

Projected dynamical systems. When boundary contact does occur, the clamping operation implements a projection onto the constraint set $\mathcal{X}$. Projected dynamical systems preserve contraction properties when the constraint set is convex Lohmiller and Slotine [1998], which $\mathcal{X}$ (a hyperrectangle) is.

Pre-emptive tightening. The saturation diagnostic of Section C monitors the I-channel's distance from the boundary. When $m_{\mathrm{sat}} < 0.1$ the system logs a warning and the adaptive governor tightens thresholds pre-emptively, preventing boundary contact rather than relying on clamping to maintain invariance.